

Term 7- Sept 2025

Nanoelectronics and Technology (01.119/99.503)-Week 9 Class 1

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11-Nov- 2025

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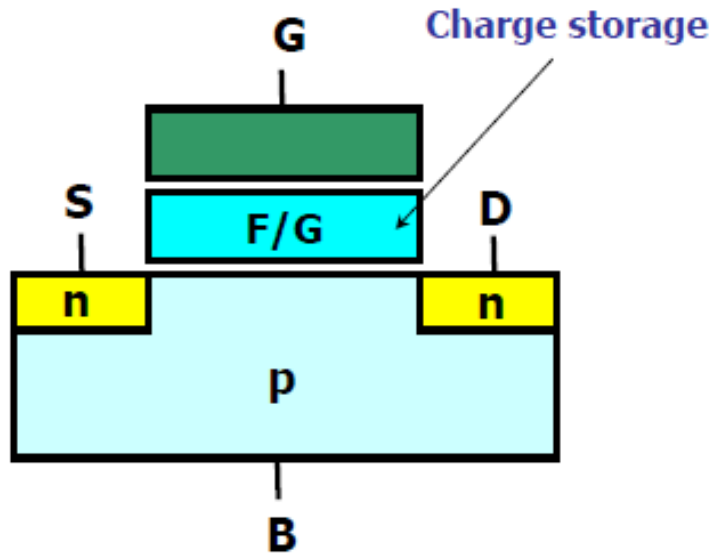
Outline

- Non-volatile Random Access Memory
 - Working principle of Flash memory
 - Challenges
- Emerging Non-volatile memory
 - RRAM
 - MRAM

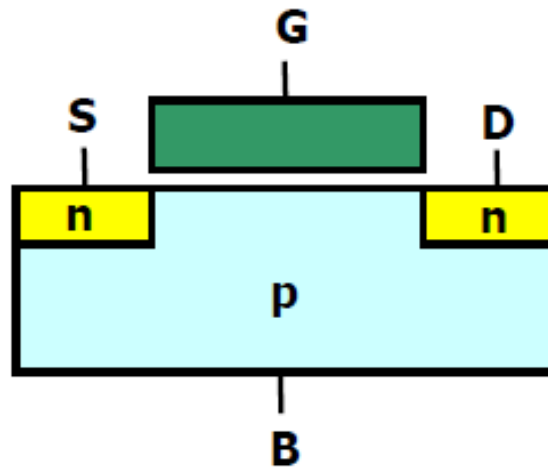
Flash memory: Non-volatile

Flash memory cell vs. MOSFET

- Flash cell has a charge storage layer such that V_{th} of a cell can be changed → memorize information



Flash cell



MOSFET

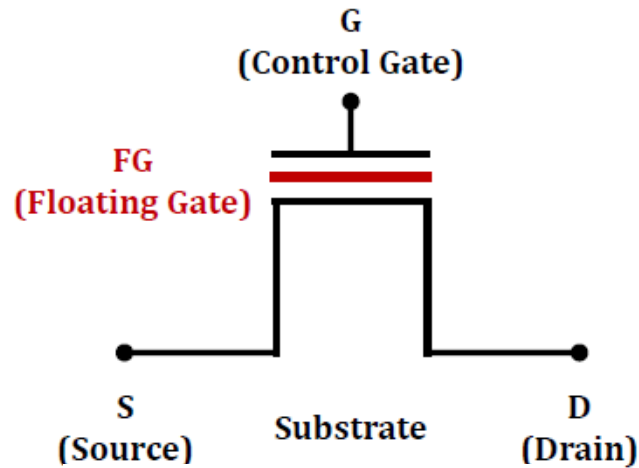


Flash Memory

K. Choi, Samsung Electronics, co., Ltd
Flash design team 'Nand flash memory' 2010.

Flash memory: Non-volatile

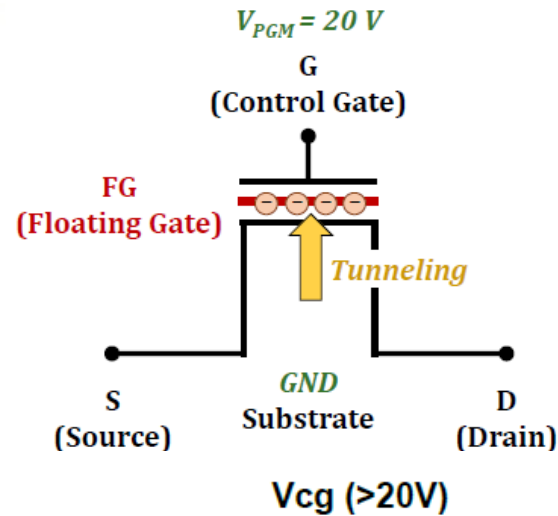
Flash memory cell



Erase : FN tunneling

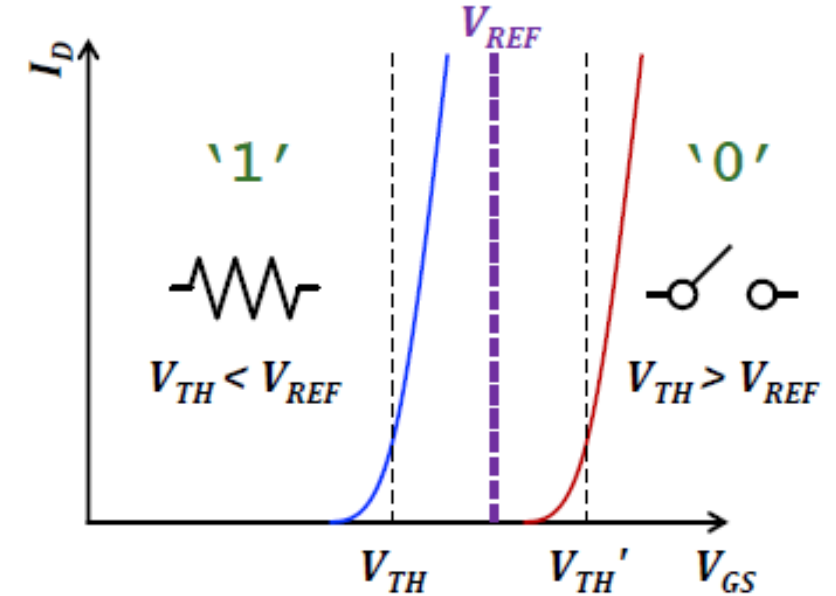
On cell $\rightarrow$ Data "1"

Jihong Kim (SNU) NAND Flash Overview.



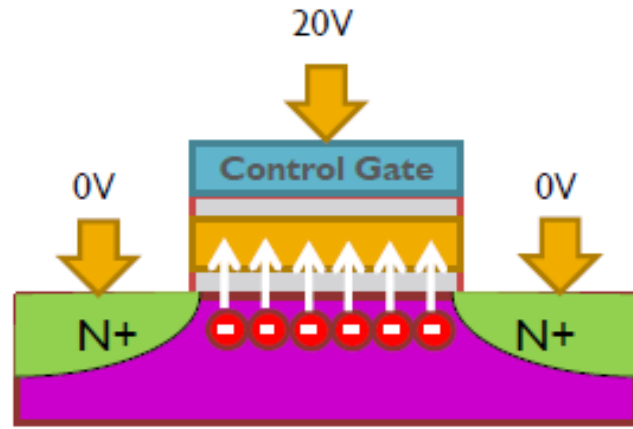
Program : FN tunneling

Off cell $\rightarrow$ Data "0"



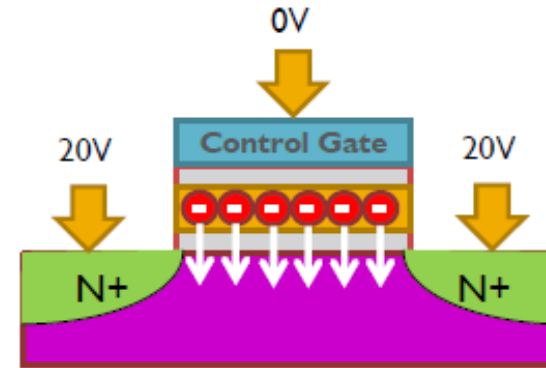
Flash memory: Non-volatile

Restricted



Program / write operation

- ▶ Apply high voltage to the control gate
- ▶ → electrons get trapped into the floating gate



Erase operation

- ▶ FN (Fowler-Nordheim) tunneling: Apply high voltage to substrate (compared to the operating voltage of the chip - usually between 7–20V)
- ▶ → electrons off the floating gate

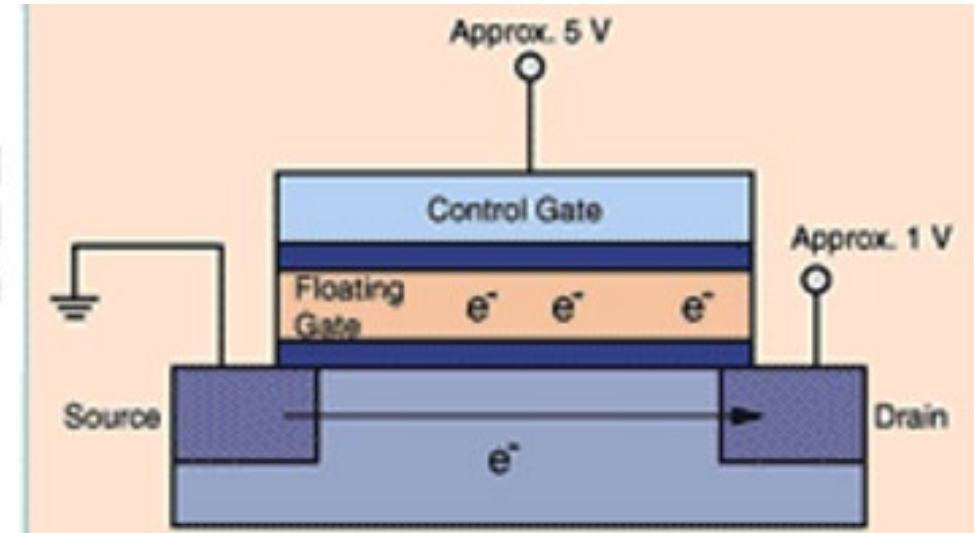
Restricted

Flash memory: Read operation

Restricted

Read

Applies the voltage at approx. 5 V to the control gate and at approx. 1 V to the drain. The state of the memory cell is distinguished by the current flowing between the drain and the source.



- If the selected cell has **low V_{th}** (logic '1'):
 - The transistor conducts → **current flows** through the NAND string.
- If the cell has **high V_{th}** (logic '0'):
 - The transistor does **not conduct** → **no current** flows.

Restricted

Flash memory:

- Faster in operation and more durable/reliable.
- Lighter and smaller in size.
- Uses less power and hence produces less heat
- Higher cost.
- Limited number of write and erase cycles

Flash memory-Challenges

Endurance

A limited number of program/erase operations

Retention

Data loss by leakage

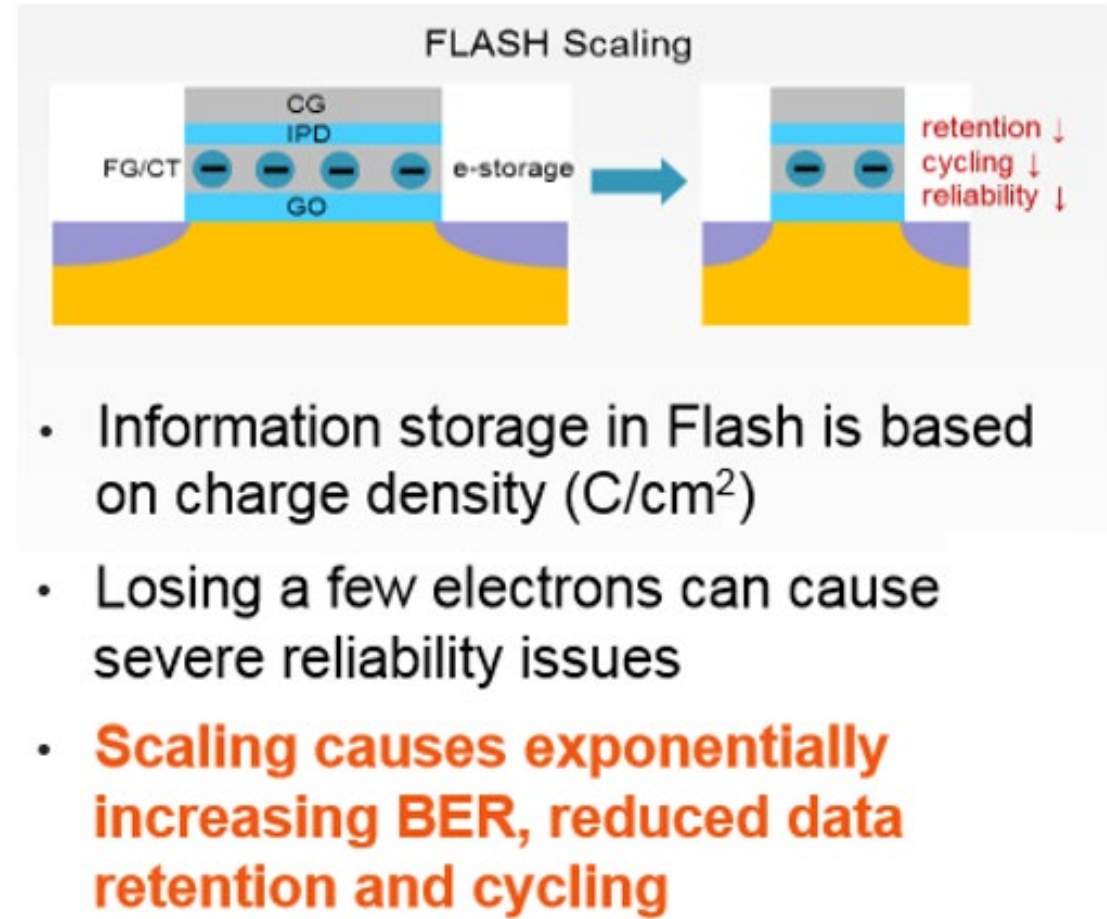
Read Disturb

A limited number of read operations

Flash memory-Challenges

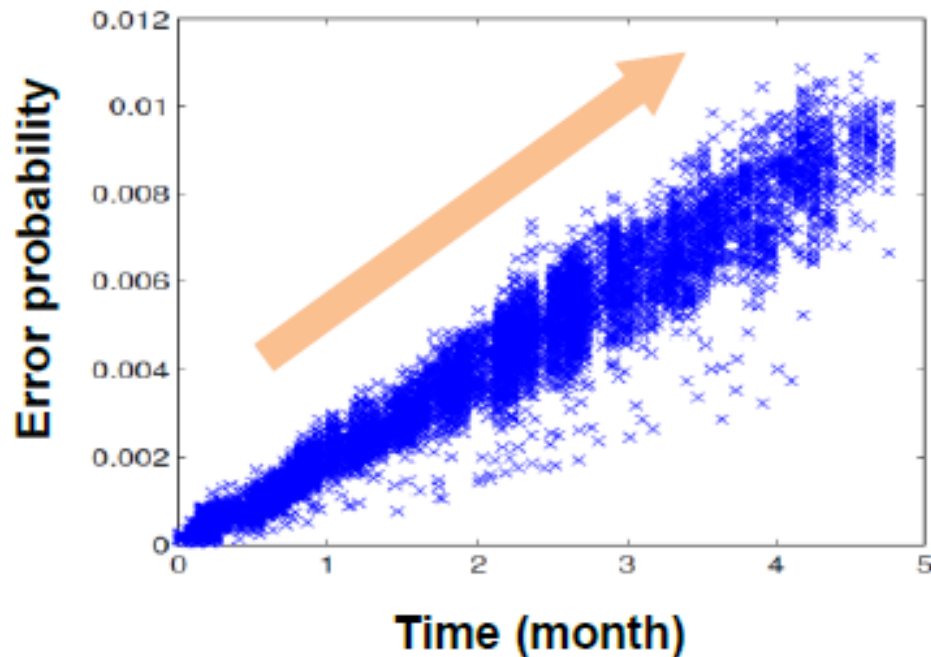
Retention:

- The **retention problem** in **NAND flash memory** refers to the gradual **loss of stored charge** in the memory cell over time, leading to **data corruption** or **bit errors** during read operations.
- This is one of the most critical reliability issues in modern NAND technology, especially as feature sizes shrink



Flash memory-Challenges

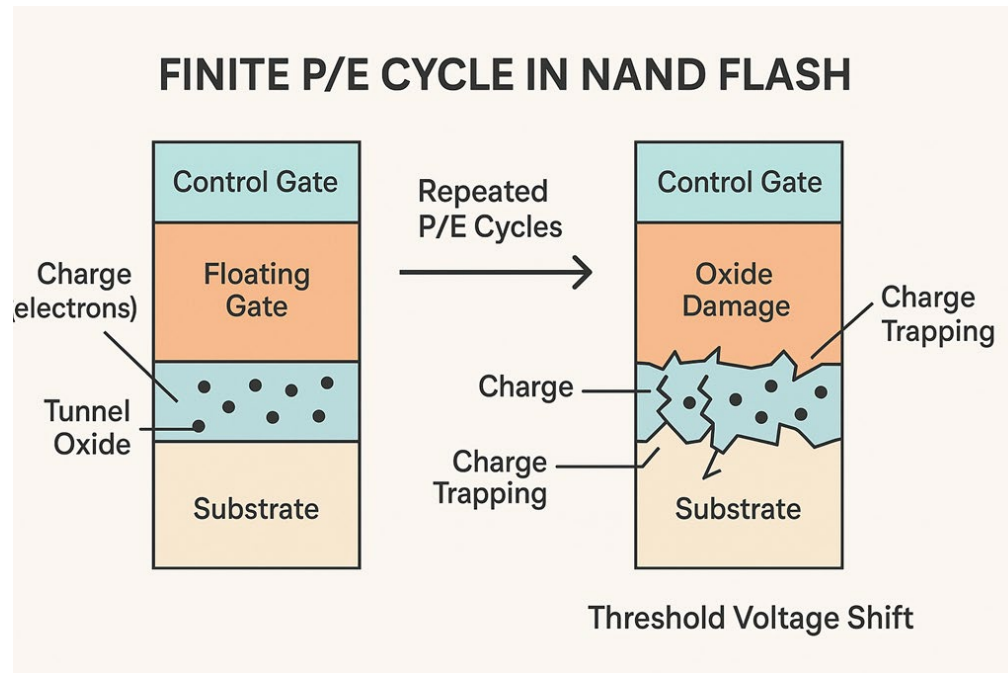
- The retention concept is the ability of a memory to keep a stored information over time with no biases applied
 - Thermal stress (= high ambient temperature) leads to charge loss which means cell V_{th} shift to lower state
 - Charge loss causes an increase in BER, eventually read failures



Jihong Kim (SNU)

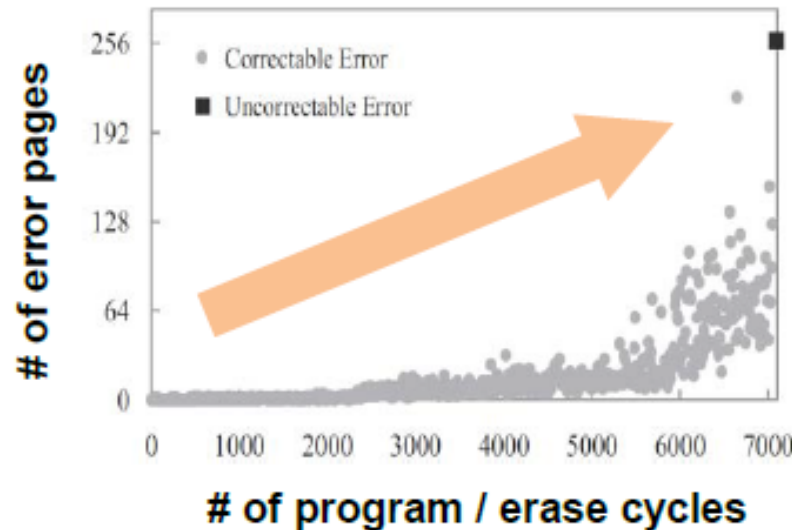
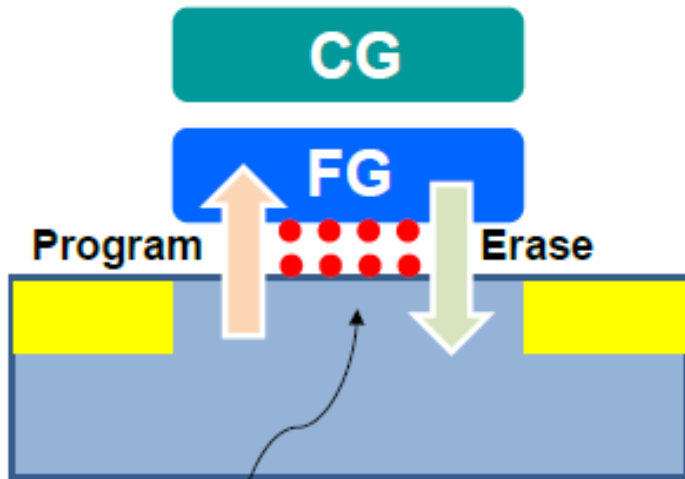
Flash memory-Challenges: P/E cycle

- **Finite P/E cycle** refers to the limited number of **Program/Erase cycles** each memory cell can endure before it wears out and becomes unreliable.
- **P/E Cycle**: One **program (write)** operation followed by one **erase** operation on a NAND flash cell.
- Each cycle stresses the **floating gate** or **charge trap** dielectric, leading to **oxide degradation** and **charge trapping**.
- After a certain number of P/E cycles, the cell can no longer reliably store charge, leading to **bit errors**.



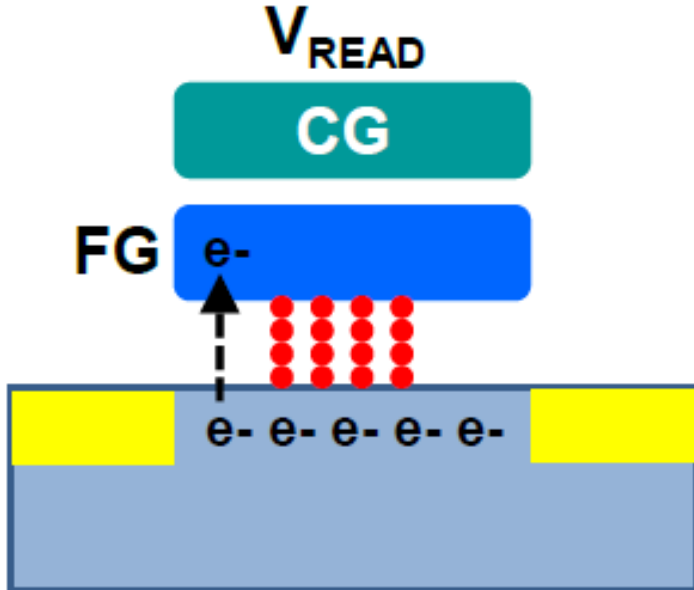
Flash memory-Challenges

- Repeated program & erase operations wear-out NAND flash because of tunnel oxide degradation
 - High voltage in program & erase operation give a damage to tunnel oxide, and then increase interface and oxide trap density
 - The larger number of P/E cycles, the higher BER (Bit error rate)
 - Tunnel oxide damage accelerates the reliability degradation, for example retention error and read disturb error



Flash memory-Challenges

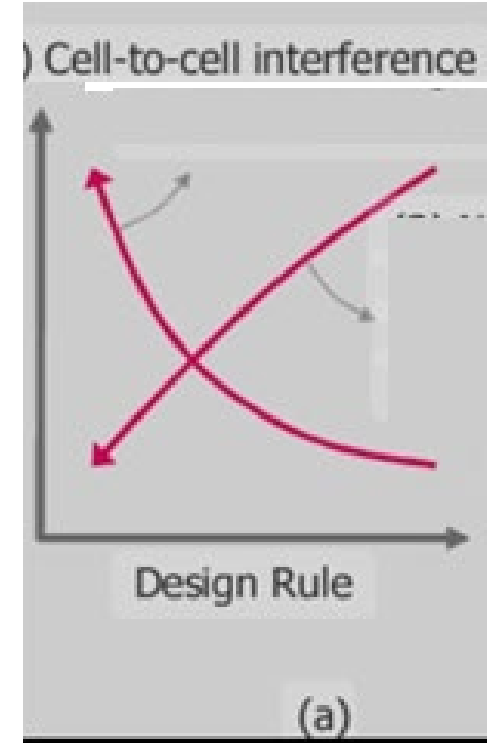
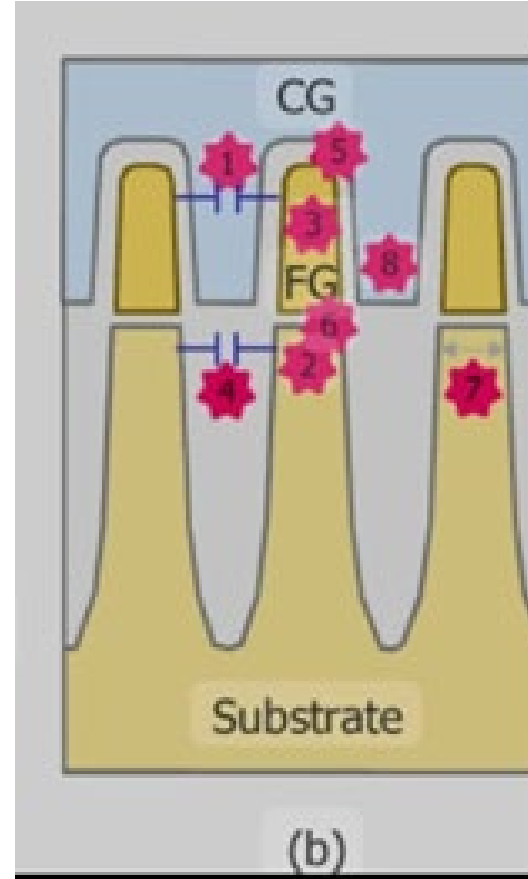
- Read disturb is an unexpected soft-program phenomenon due to repeated read operations
 - V_{READ} (6~7V) which is applied to unselect W/L's in read operation can cause a soft-programming.
 - Read stress increases with the read voltage and read operation time
 - As P/E cycles increase, read disturb errors are accelerated



Flash memory-Challenges

Cell to cell interference effect

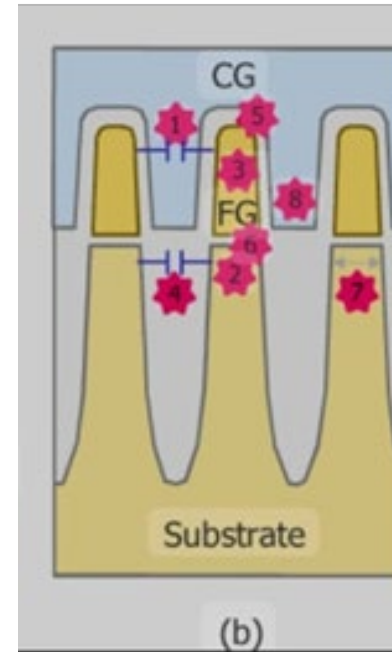
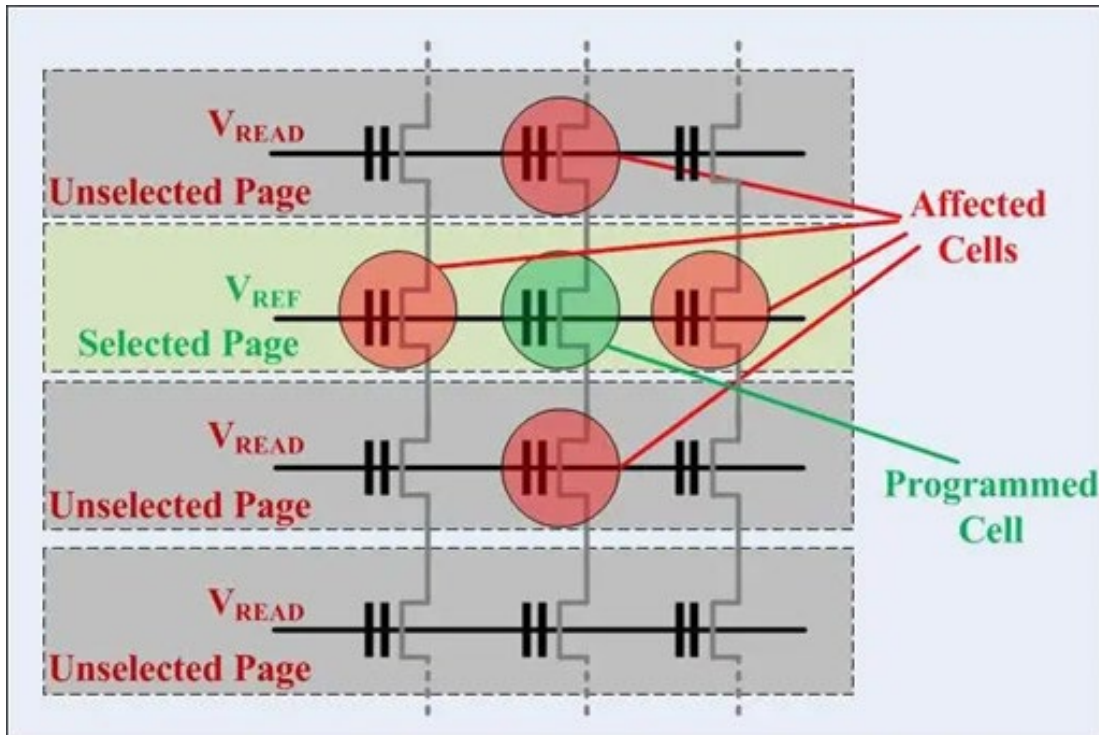
- Cell-to-cell interference (also called **cell-to-cell coupling** or **program interference**) in NAND flash memory arises because memory cells are packed **very closely together** and **share electric fields** through their floating gates or charge-trap layers.
- When one cell is programmed, nearby cells experience **unintended threshold-voltage shifts**, leading to data corruption over time.



Flash memory-Challenges

Cell to cell interference effect

- Neighboring cells (both on the same word line and adjacent word lines) experience **coupling capacitance** through:
 - **Gate-to-gate coupling** (between adjacent floating gates)
 - **Gate-to-channel coupling**



Flash memory-Challenges

Endurance

A limited number of program/erase operations

Retention

Data loss by leakage

Read Disturb

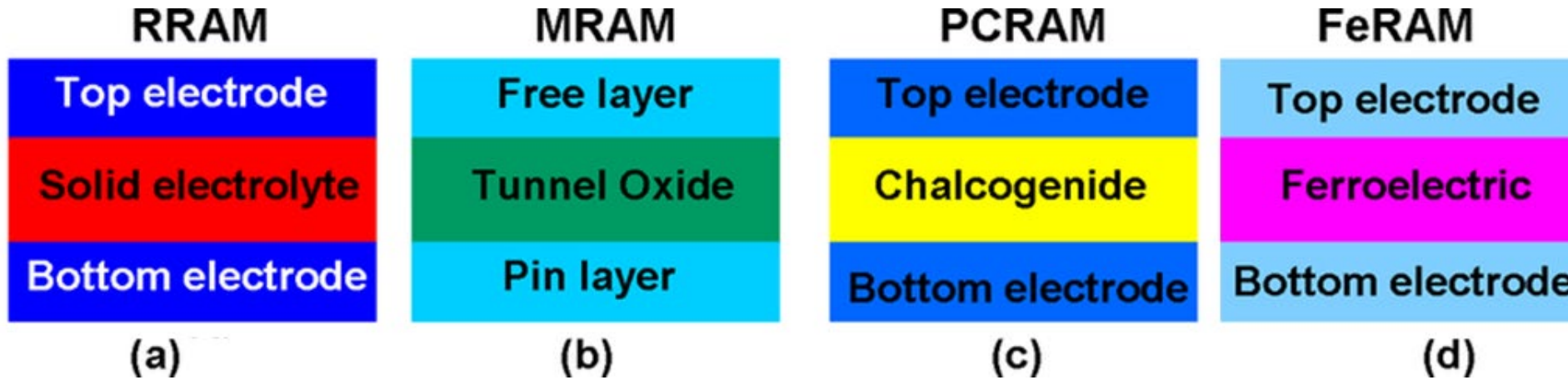
A limited number of read operations

Flash memory-Challenges

- Planar NAND is hitting physical limits at about the 1z nm node: The cell size cannot be reduced much further.
- 3D NAND relaxes the cell dimensions and adds additional layers in the vertical direction.
- 3D NAND appears to be the only viable option to increase densities until a better non-volatile mass storage technology emerges.

NV-Memory: Future

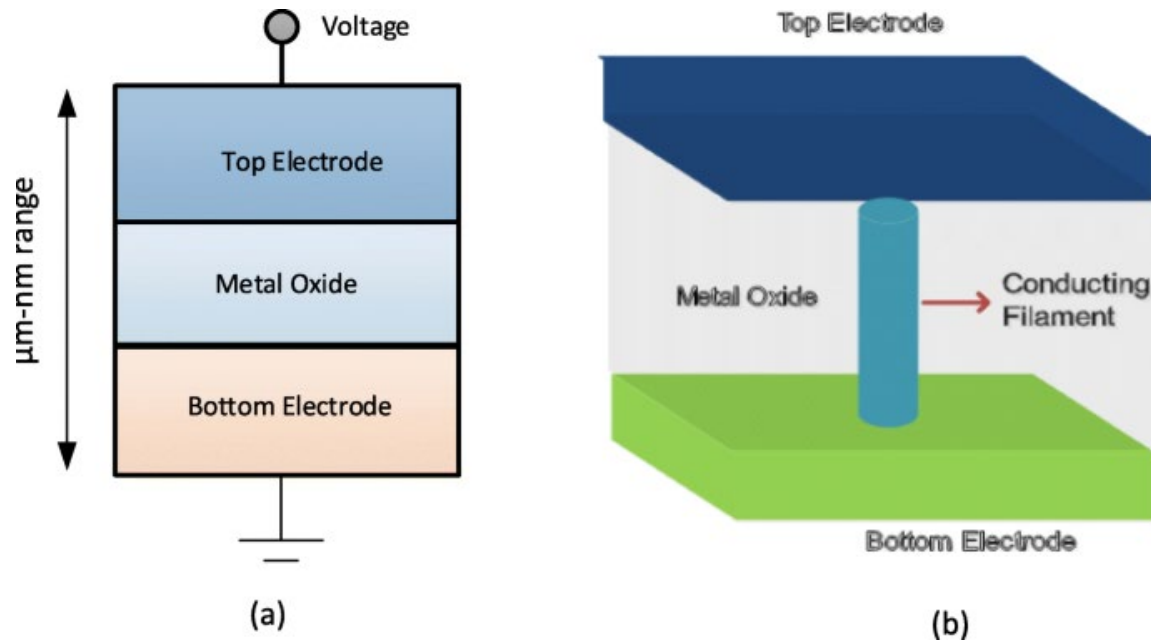
Restricted



Novel Memory



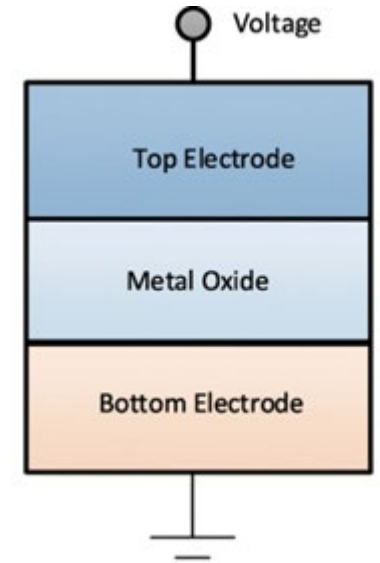
Resistive RAM (RRAM or ReRAM)



Resistive random-access memory (ReRAM or RRAM) is a type of non-volatile (NV) random-access (RAM) computer memory that works by changing the resistance across a dielectric solid-state material, often referred to as a memristor.

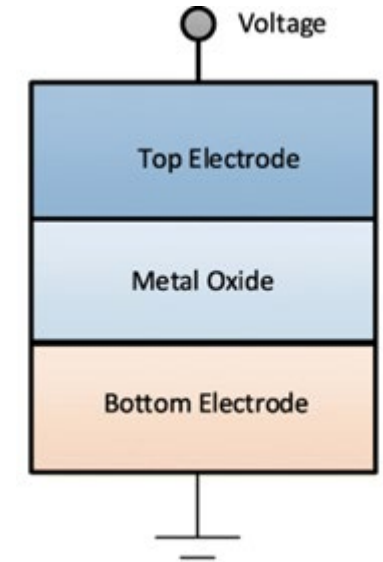
RRAM

- ReRAM bears some similarities to [conductive-bridging RAM \(CBRAM\)](#), CBRAM involves one electrode providing ions that dissolve readily in an electrolyte material.
- By contrast, ReRAM involves generating defects in a thin oxide layer, known as oxygen vacancies (oxide bond locations where the oxygen has been removed), which can subsequently charge and drift under an electric field. The motion of oxygen ions and vacancies in the oxide would be analogous to the motion of electrons and holes in a semiconductor.



RRAM

- Although ReRAM was initially seen as a replacement technology for **flash memory**, the cost and performance benefits of ReRAM have not been enough for companies to proceed with the replacement.
- Apparently, a broad range of materials can be used for ReRAM. However, the discovery that the popular high-k gate dielectric **HfO₂** can be used as a low-voltage ReRAM has encouraged researchers to investigate more possibilities.

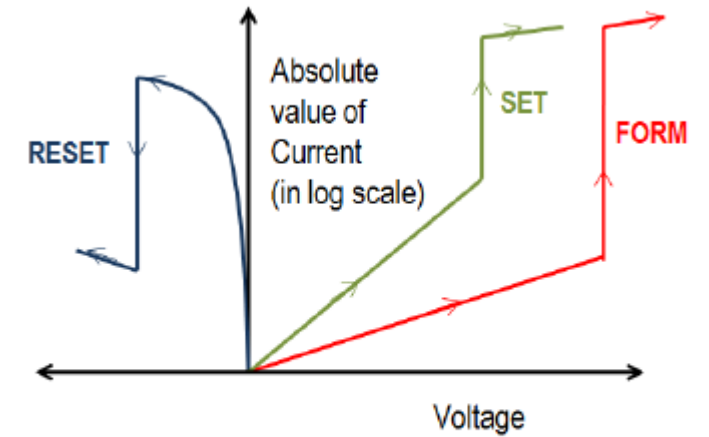
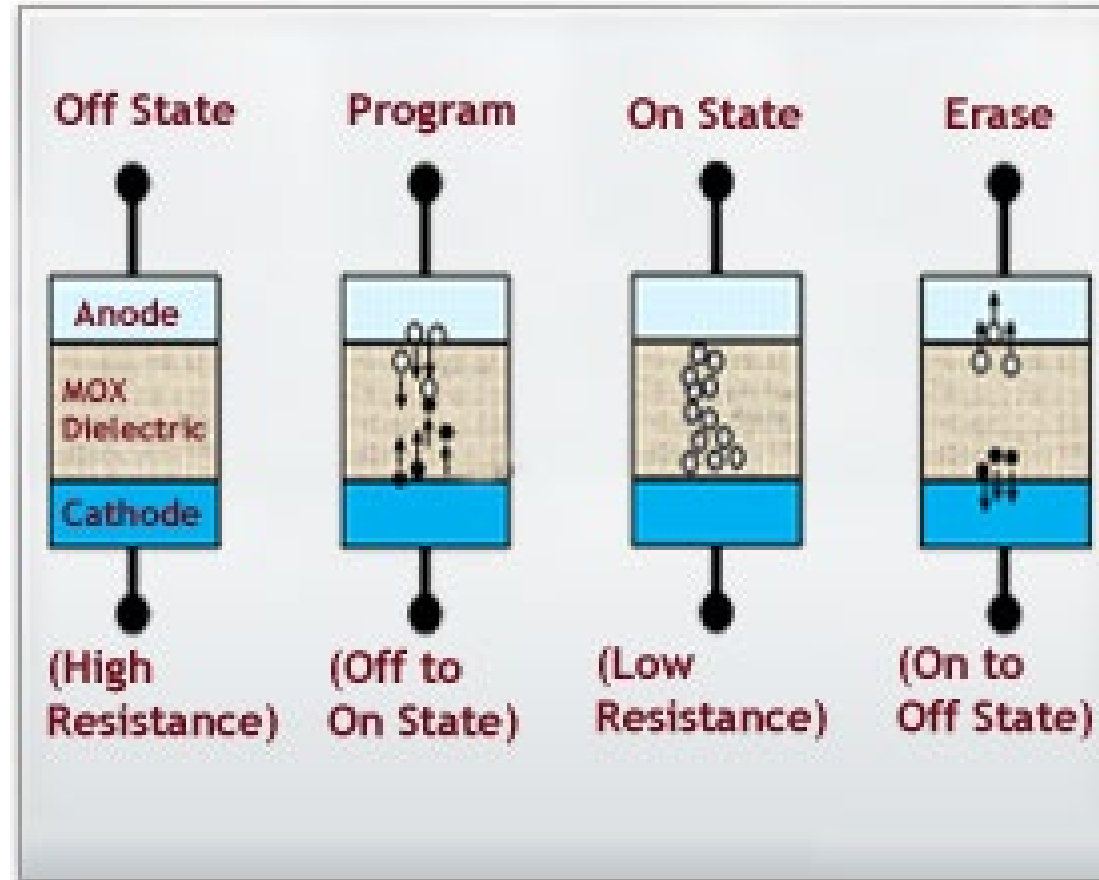


Top electrode
Transition Metal Oxide
Bottom electrode

	Examples
Top electrode	Pt, TiN/Ti, TiN, Ru, Ni ...
Transition Metal Oxide	TiO _x , NiO _x , HfO _x , WO _x , TaO _x , VO _x , CuO _x , ...
Bottom Electrode	TiN, TaN, W, Pt, ...

RRAM

Restricted

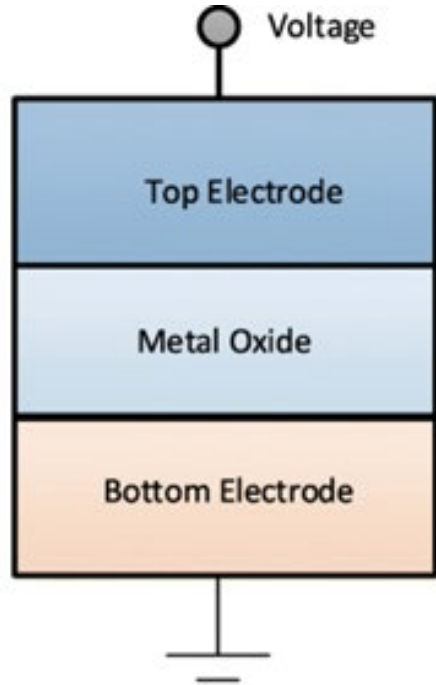


Bipolar switching:

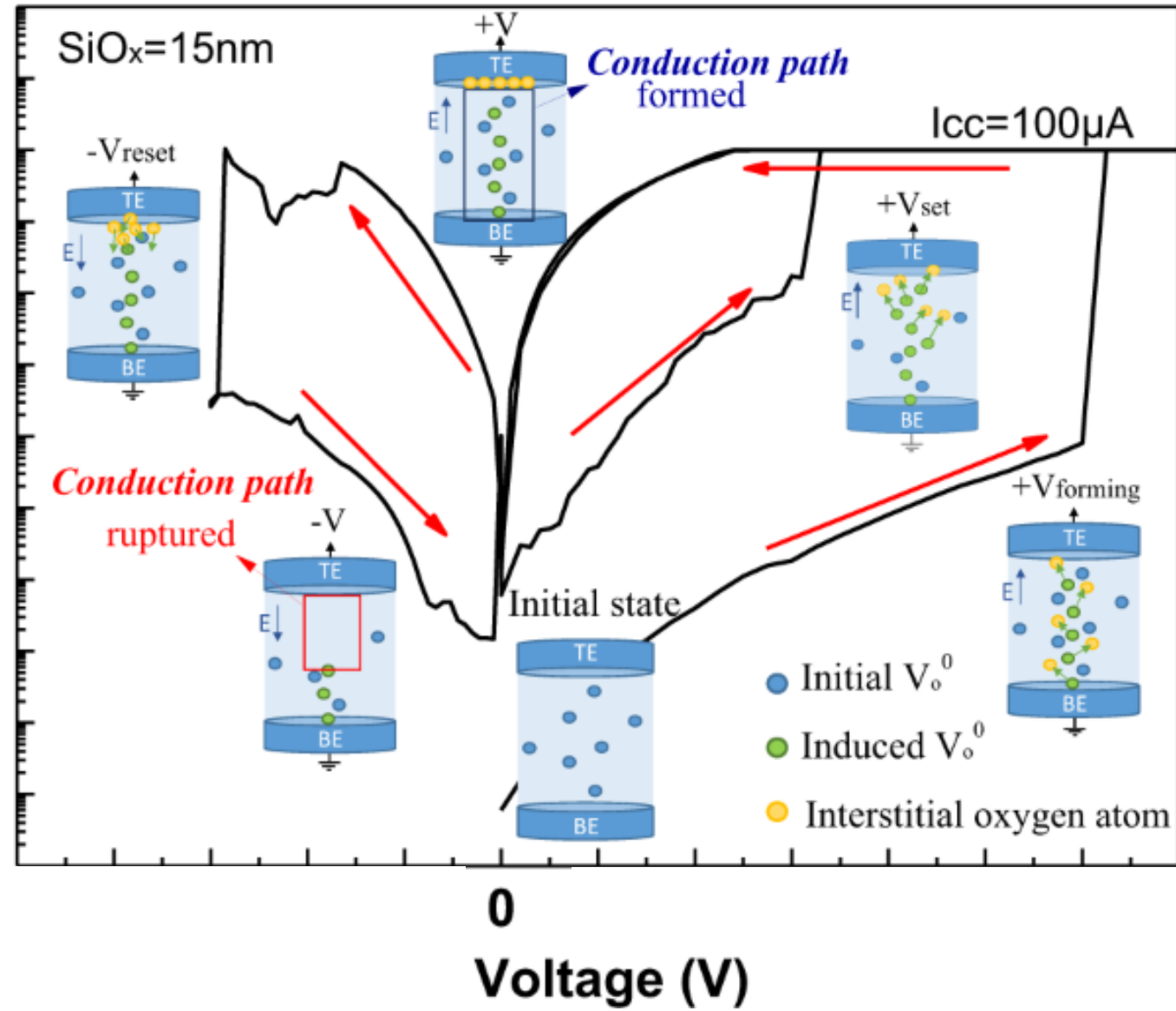
RESET opposite polarity to SET and FORM

Restricted

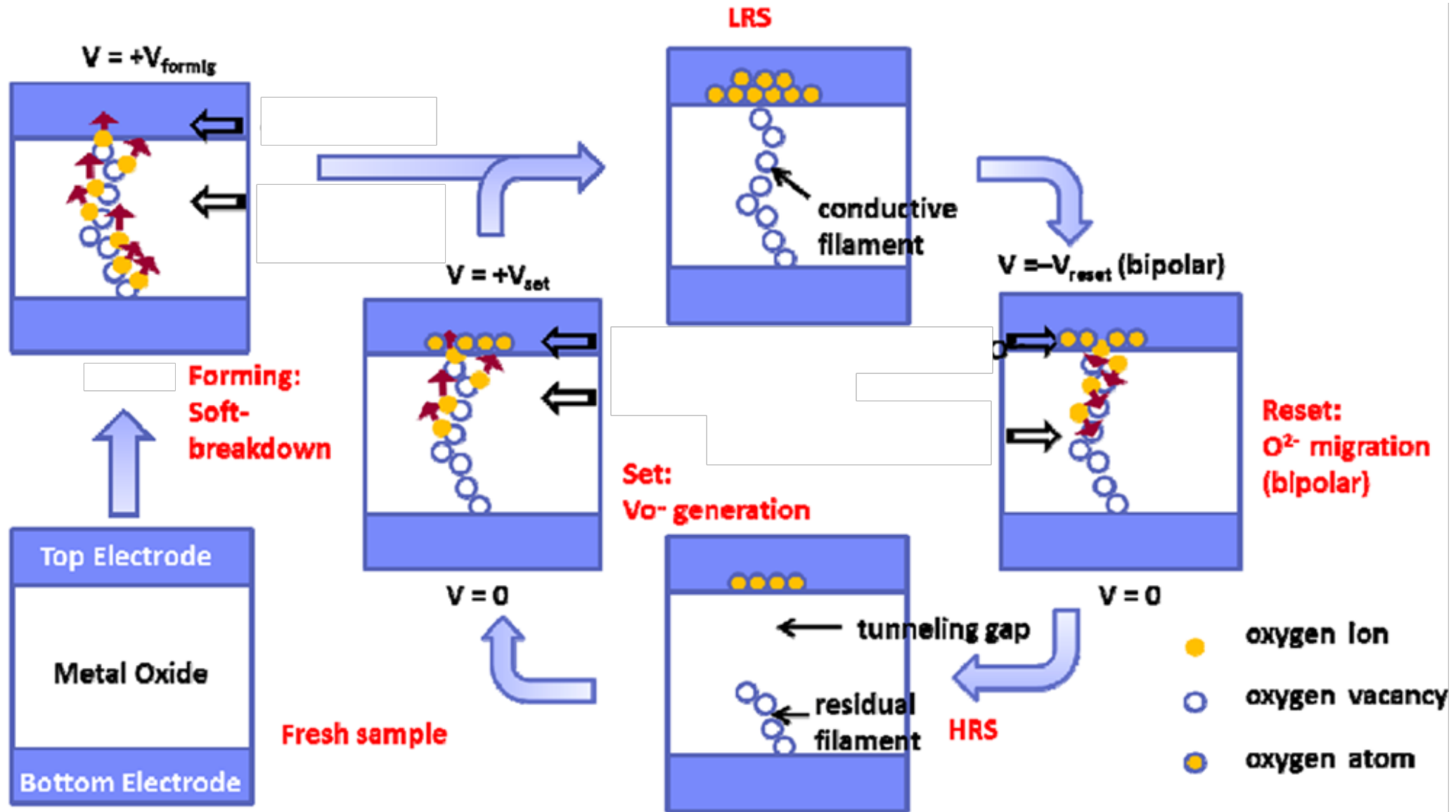
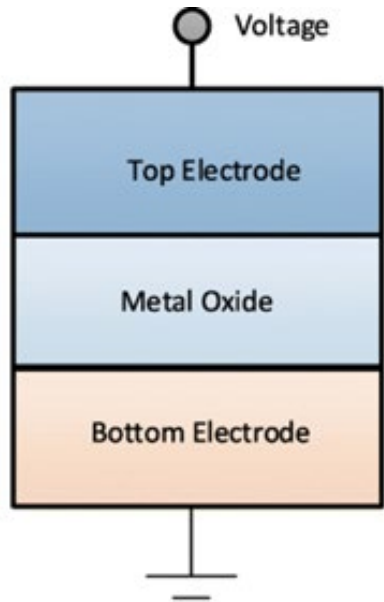
RRAM



Current (A)

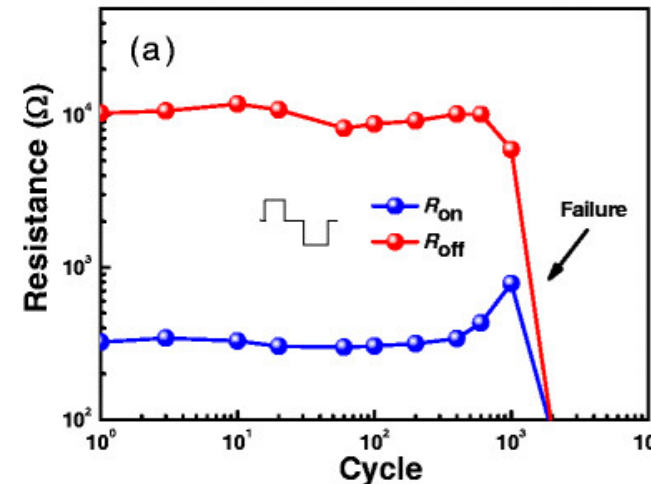
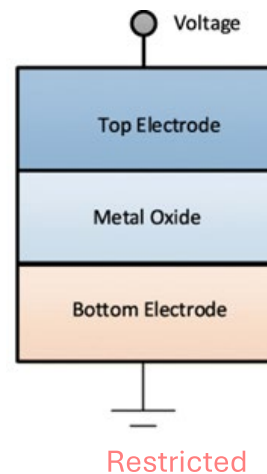
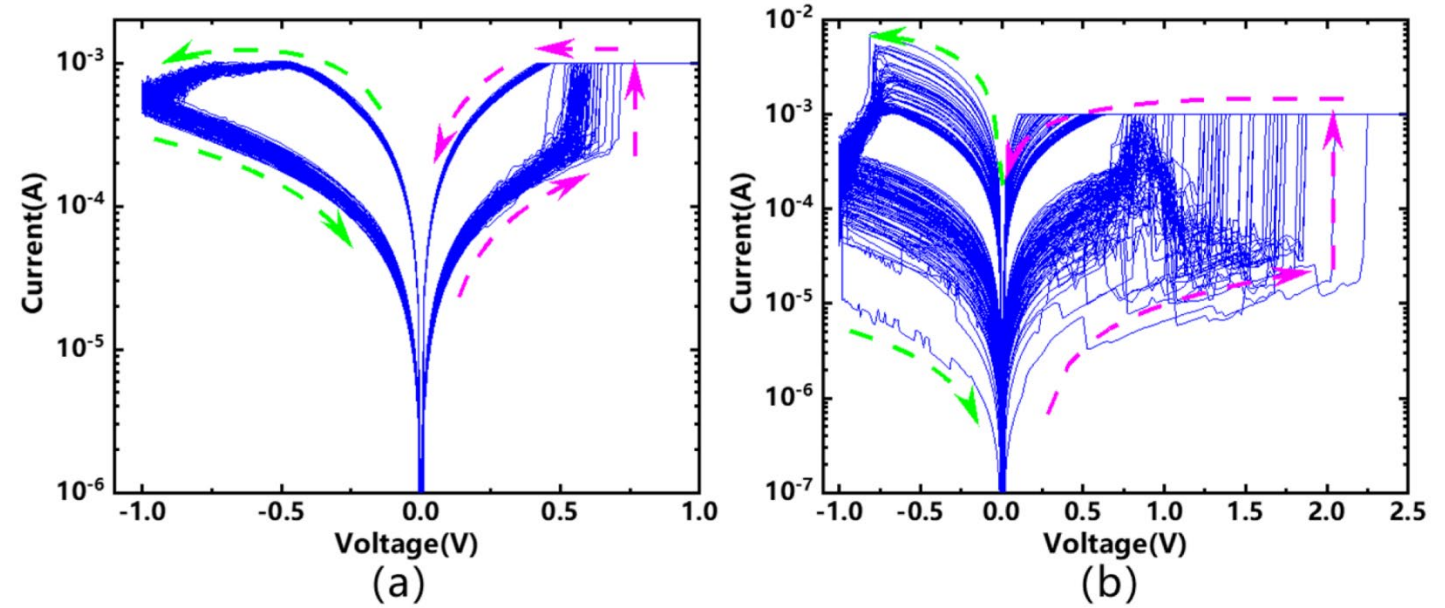


RRAM



RRAM-Endurance

- Resistive random access memory involves frequent transitions between a high resistance state (HRS) and low resistance state (LRS).
- Each switching event between the resistive states can introduce permanent damage and cause degradation of the RRAM performance.
- Endurance is thus defined as the number of times a RRAM device can be switched between the HRS and the LRS while ensuring a reliably distinguishable ratio between them

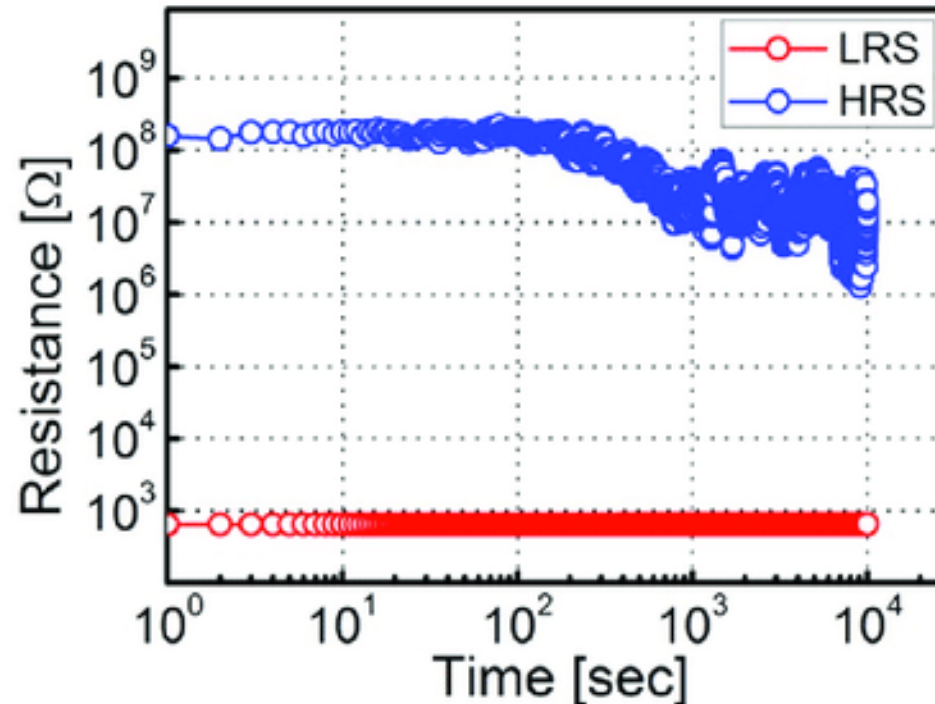
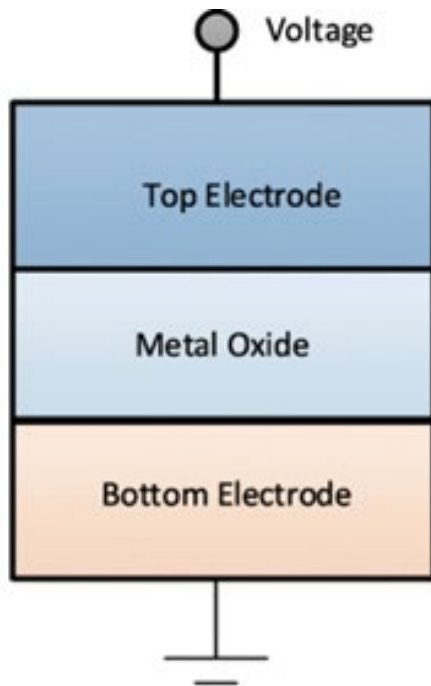


RRAM-Retention

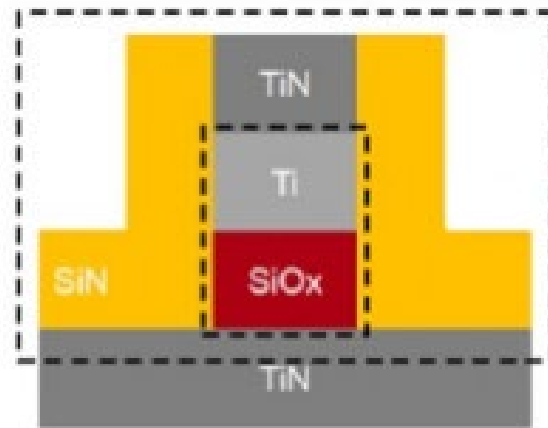
Data retention is also crucial for the practical applications of the RRAMs. To investigate the retention reliability of the RRAMs

Retention failure should be associated with redistribution of oxygen vacancies to affect electron transport and device resistance.

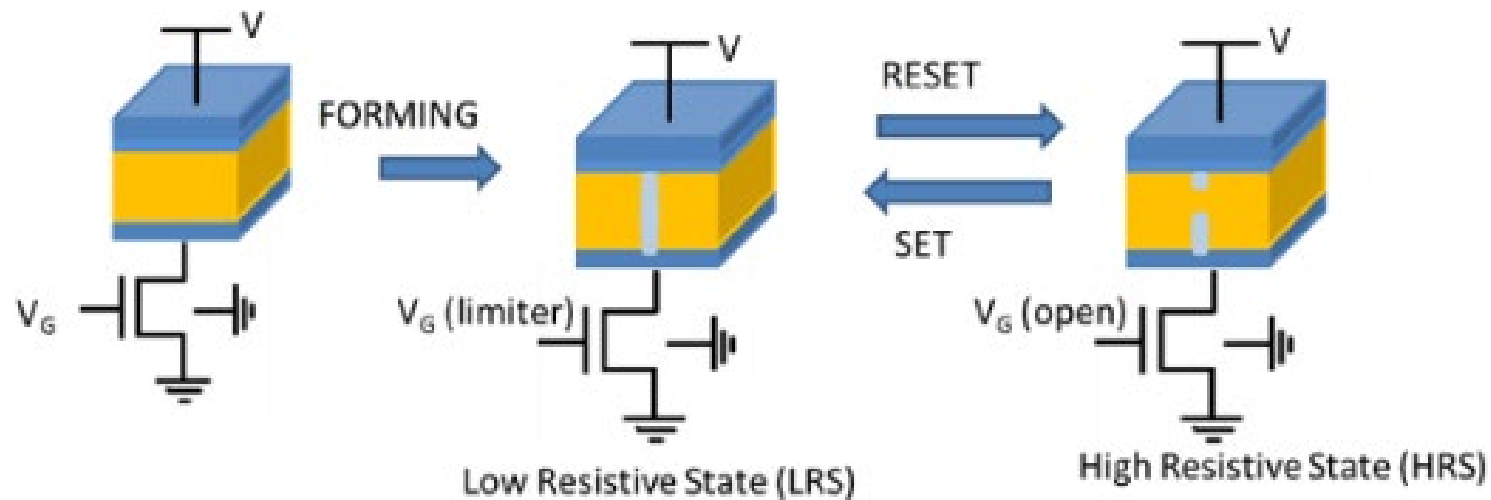
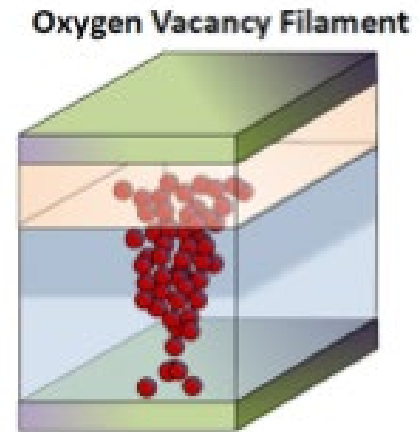
The double layers dielectric stack could hold a better HRS data retention than a single dielectric layer.



RRAM

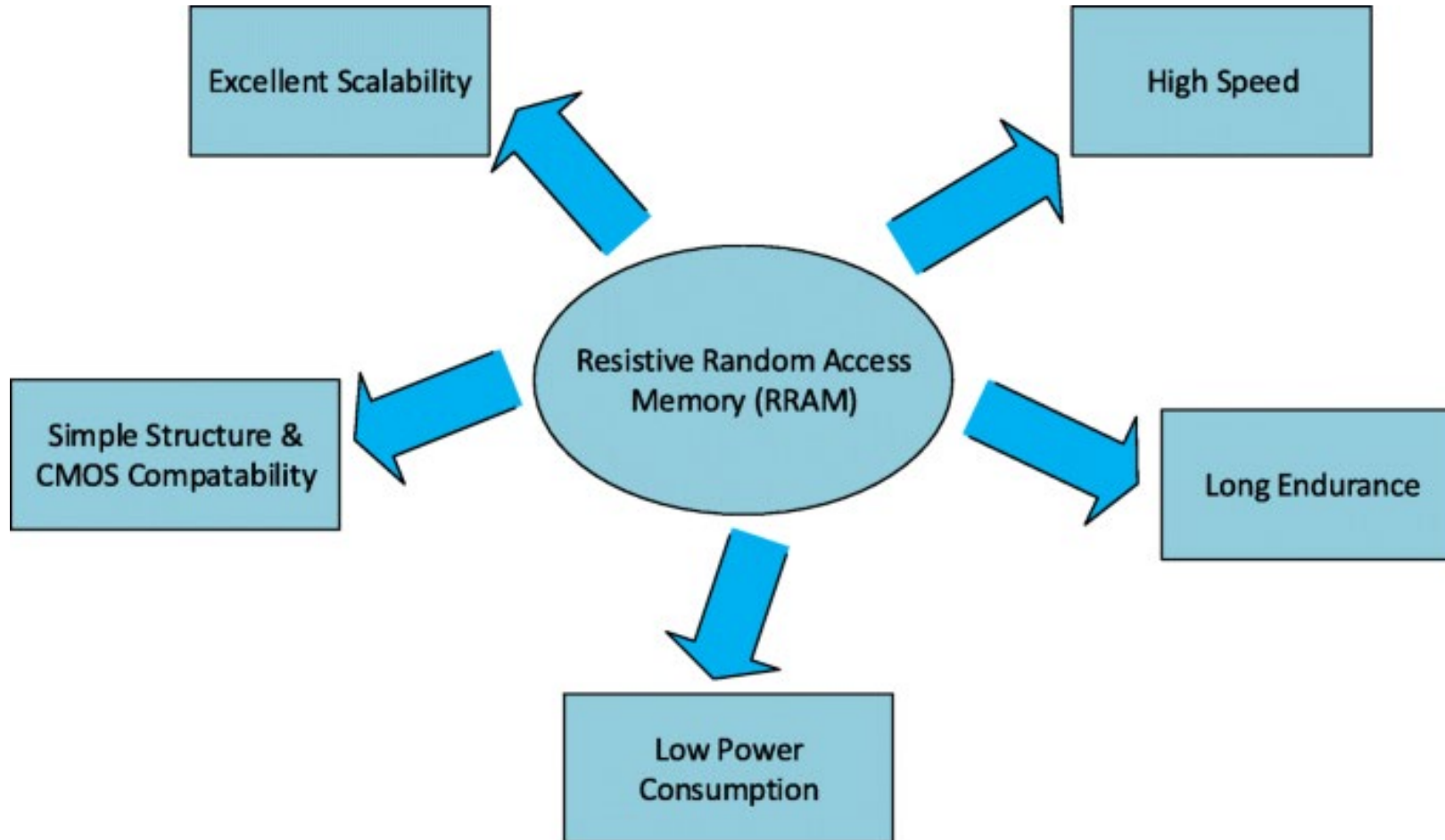


ReRAM memory



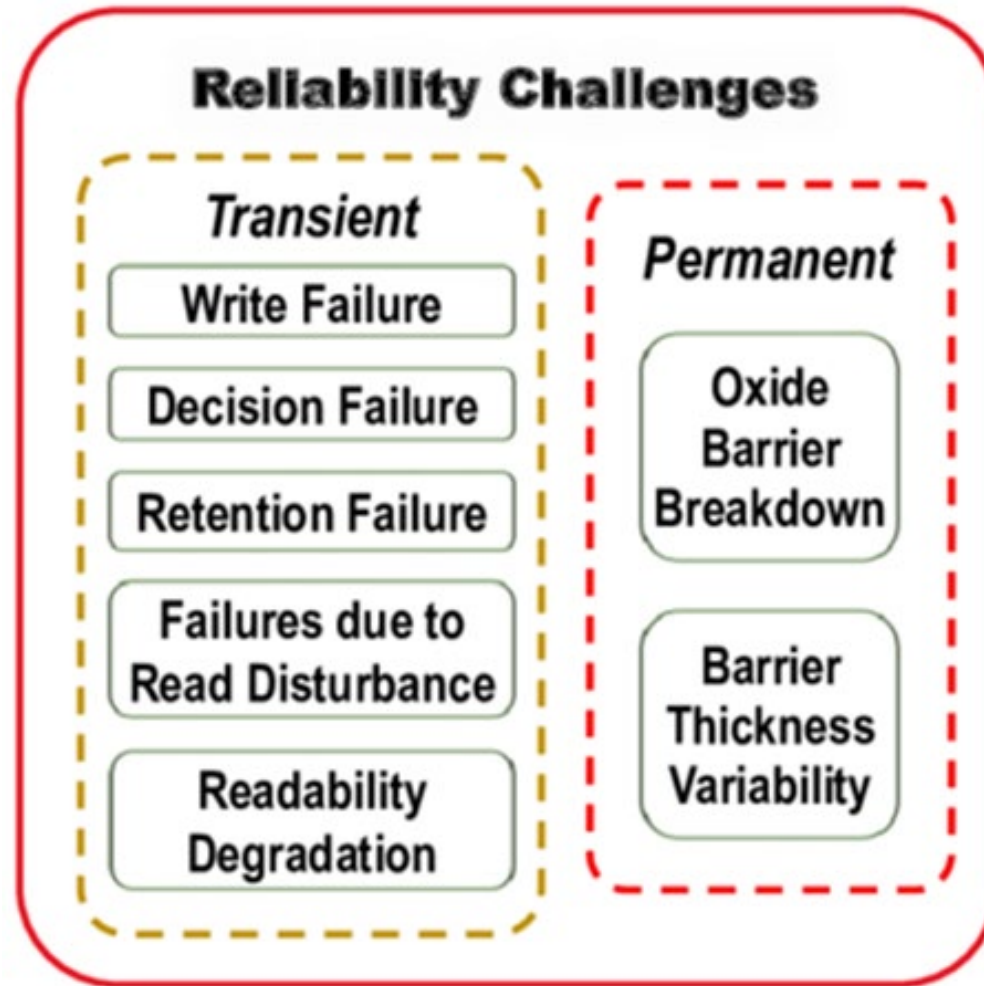
RRAM

Advantages:



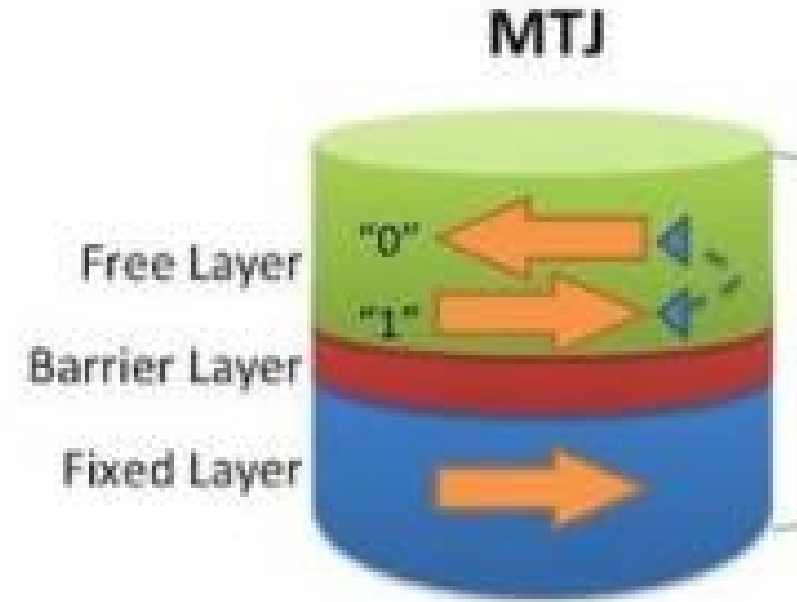
RRAM

Challenges:



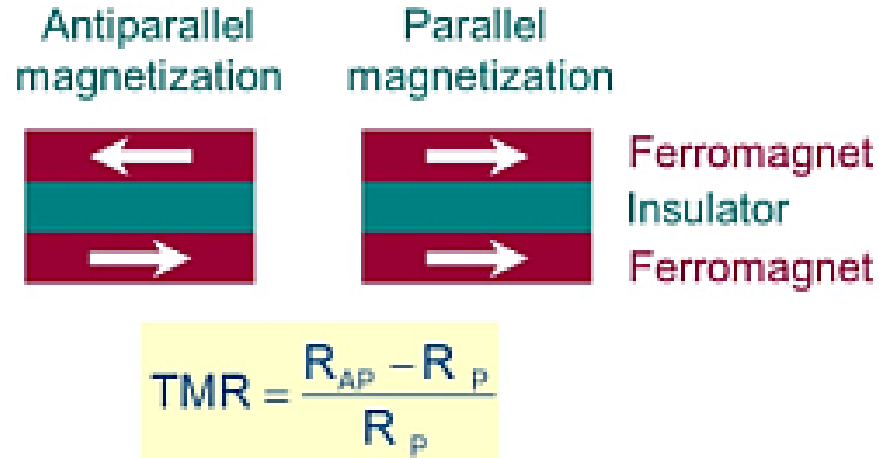
Magnetic RAM (MRAM)

- **Magnetic RAM (MRAM)** — or **Magnetoresistive Random Access Memory** — is a type of **non-volatile memory** that stores data using **magnetic states** instead of electric charge (as in DRAM or Flash).
- It aims to combine the **speed of SRAM**, the **density of DRAM**, and the **non-volatility of Flash**.



Magnetic RAM (MRAM)

Magneto-resistive random-access memory (**MRAM**) is a type of non-volatile random-access memory which stores data in magnetic domains.

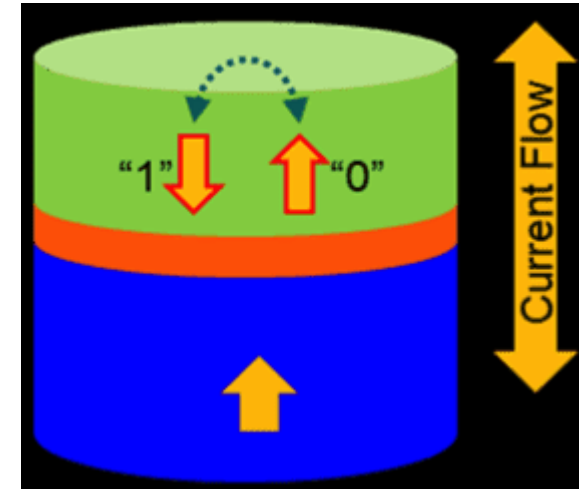


Tunnel magnetoresistance (TMR) is a magnetoresistive effect that occurs in a magnetic tunnel junction (MTJ), which is a component consisting of two ferromagnets separated by a thin insulator.

Magnetic RAM (MRAM)

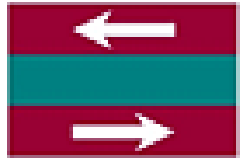
Spin Transfer Torque-MRAM (also called STT-RAM or sometimes ST-MRAM) is **an advanced type of MRAM devices**. STT-MRAM enables higher densities, low power consumption and reduced cost compared to regular devices.

- The spin of the electrons is flipped using a spin-polarized current.
- This effect is achieved in a magnetic tunnel junction (MTJ).
- A spin-polarized current is created by passing a current through a thin magnetic layer.
- This current is then directed into a thinner magnetic layer which transfers the angular momentum to the thin layer which changes its spin.

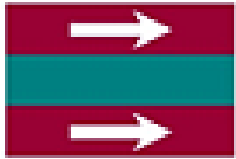


Magnetic RAM (MRAM)

Antiparallel
magnetization



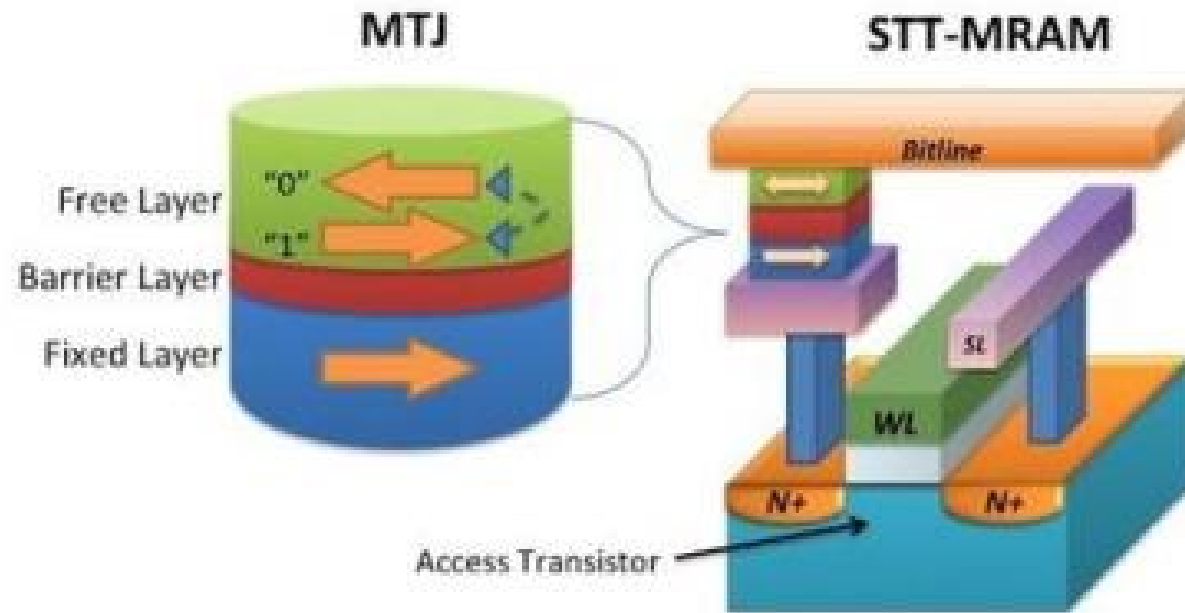
Parallel
magnetization



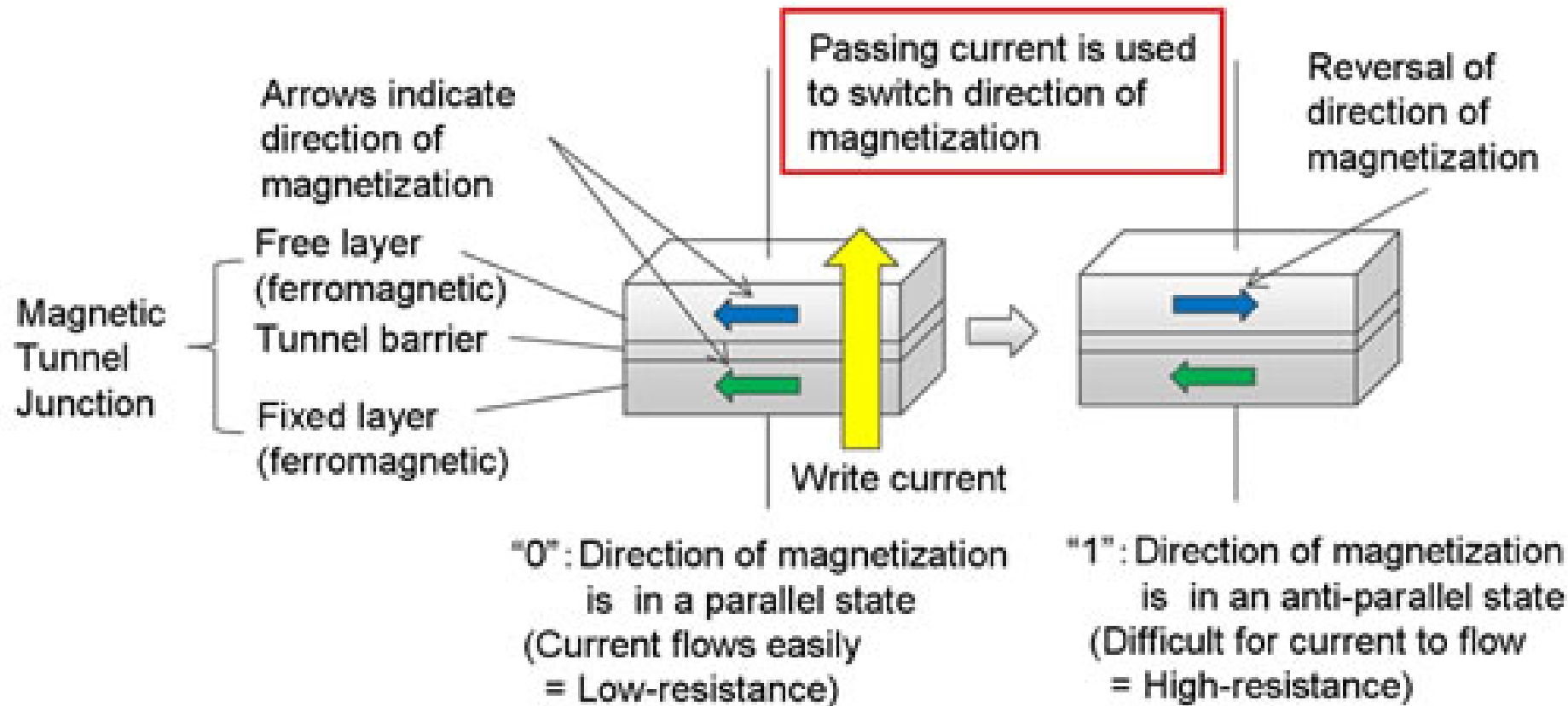
Ferromagnet
Insulator
Ferromagnet

$$\text{TMR} = \frac{R_{\text{AP}} - R_{\text{P}}}{R_{\text{P}}}$$

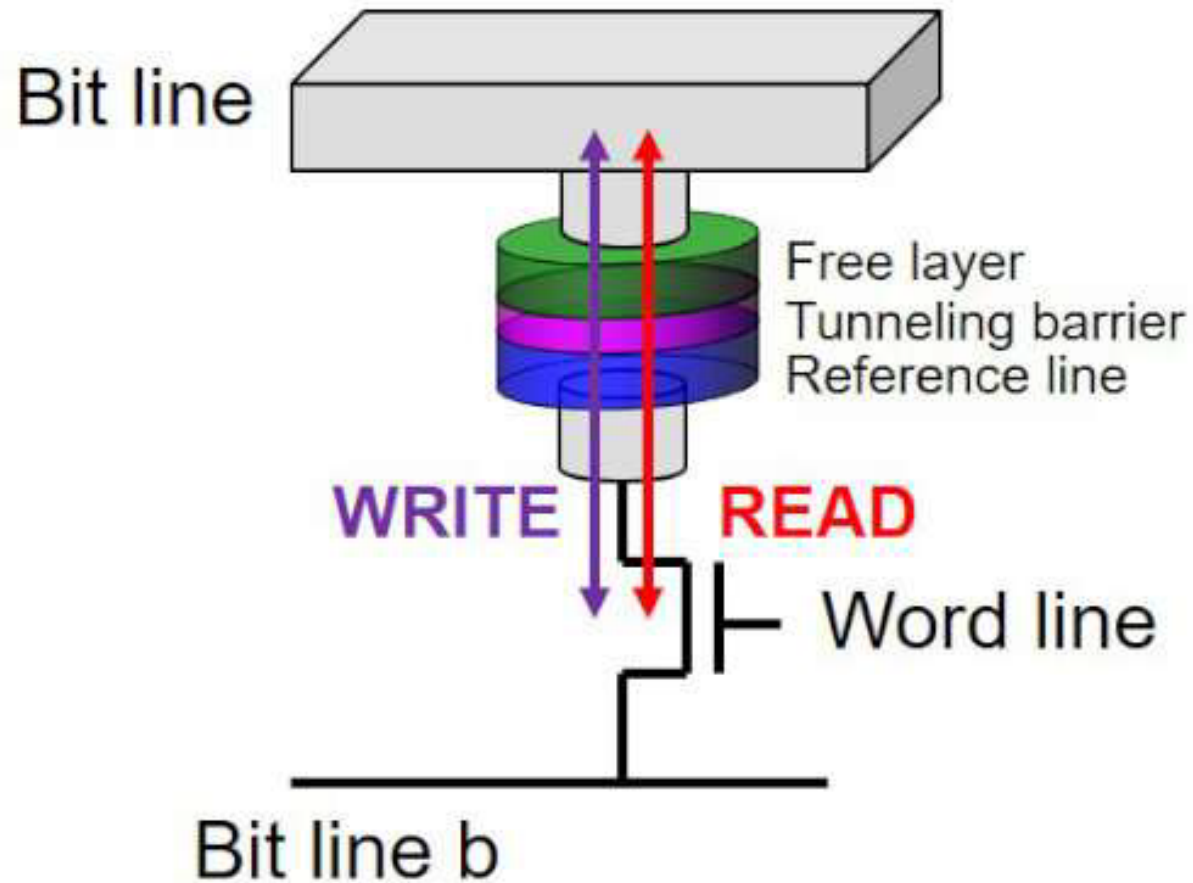
- **Fixed (reference) ferromagnetic layer** — magnetization is fixed.
- **Insulating barrier** — typically MgO (very thin, ~1 nm), allows quantum tunneling.
- **Free ferromagnetic layer** — magnetization can switch direction.



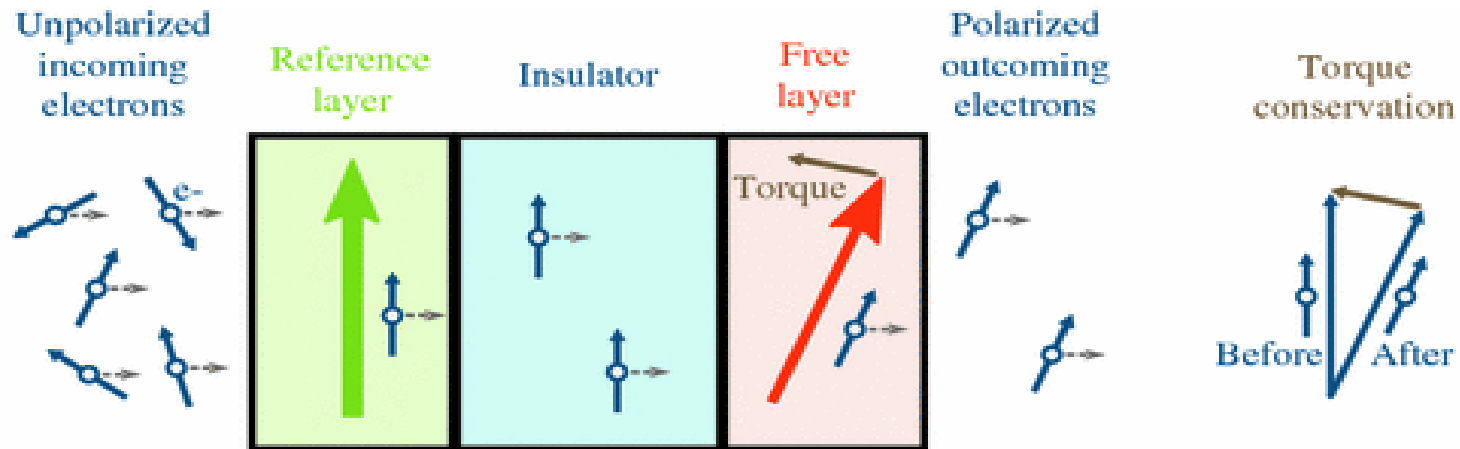
Magnetic RAM (MRAM)



Magnetic RAM (MRAM) Write/Read



Magnetic RAM (MRAM)-Write



(a) Spin-transfer torque on the free ferromagnetic layer, due to the crossing electrons (positive current).

(Electrons flow from reference → free layer)

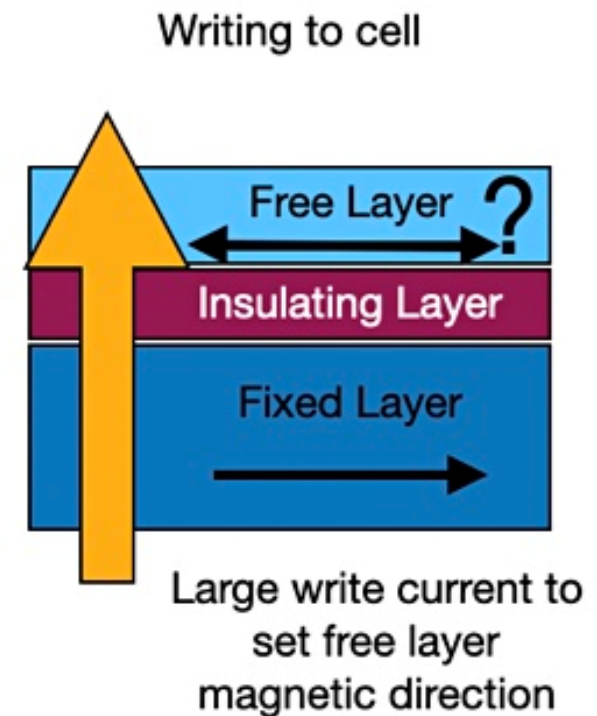
Step-by-Step Process

1. Incoming electrons:

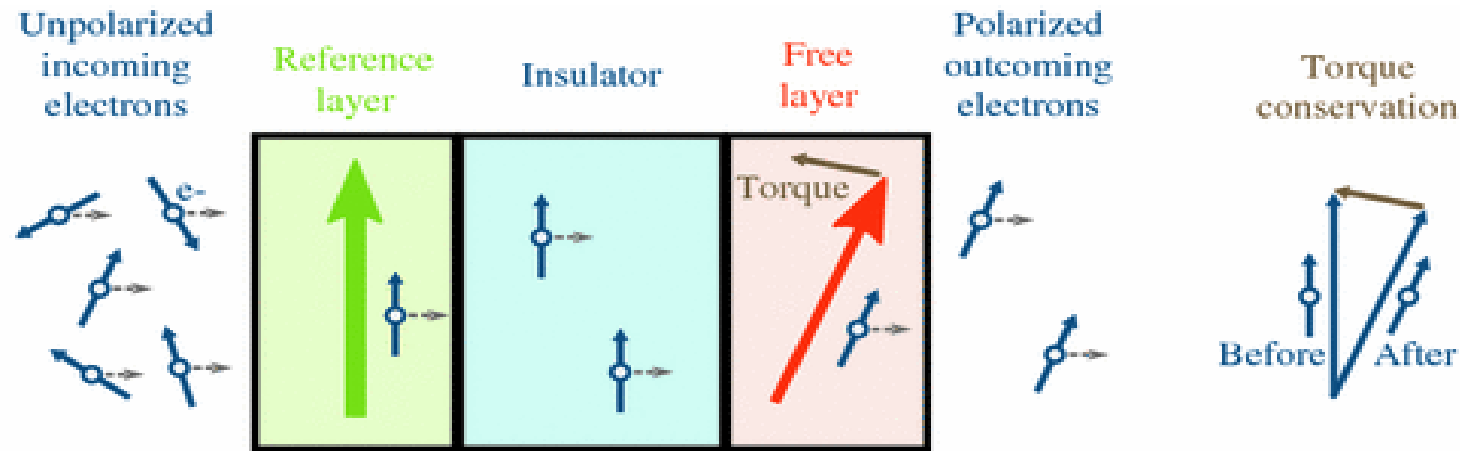
- Unpolarized electrons (blue) enter from the left.
- When they pass through the **reference layer**, only spins aligned with the fixed magnetization can transmit easily.
- Thus, the outgoing electrons become **spin-polarized** (blue arrows aligned upward).

2. Spin filtering through the barrier:

- These polarized electrons tunnel through the **insulating layer (MgO)**.



Magnetic RAM (MRAM)-Write



(a) Spin-transfer torque on the free ferromagnetic layer, due to the crossing electrons (positive current).

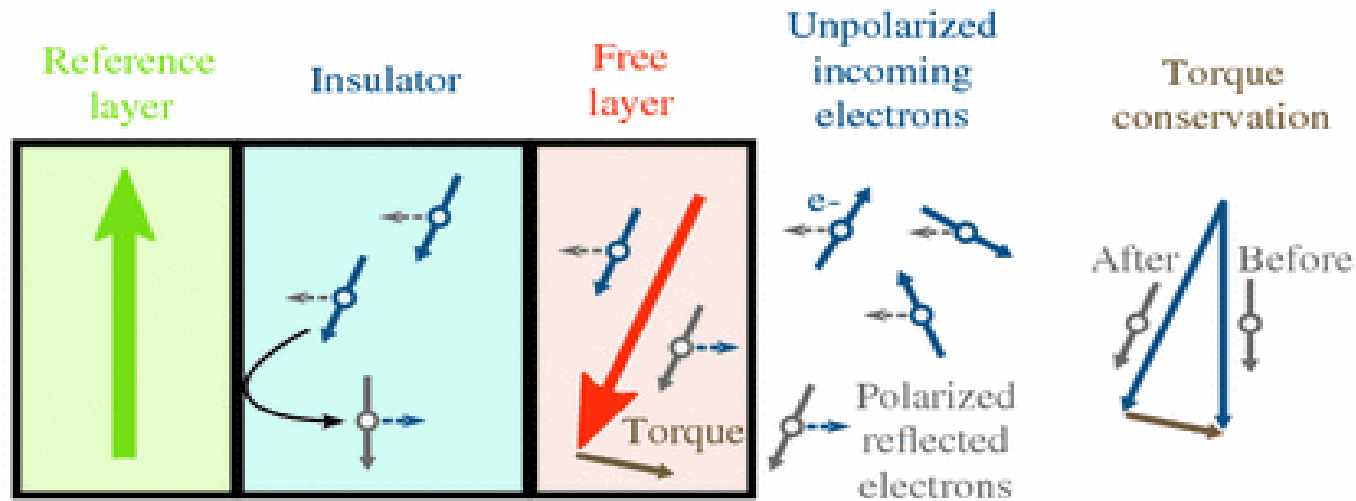
3. Interaction in the free layer:

- The electron spins entering the free layer are not aligned with the free layer's magnetization (red arrow).
- To align, the electrons transfer part of their angular momentum to the free layer.
- This transfer appears as a **spin-transfer torque (STT)** acting on the magnetization of the free layer.

4. Torque conservation:

- The total angular momentum is conserved.
- The electron spins tilt (as shown on the right) while the free layer magnetization experiences a torque trying to rotate it **toward** the direction of the incoming spin polarization.

Magnetic RAM (MRAM)-Write



(b) Spin-transfer torque on the free ferromagnetic layer, due to the reflected electrons (negative current).

(Electrons flow from free layer $\rightarrow$ reference layer)

Step-by-Step Process

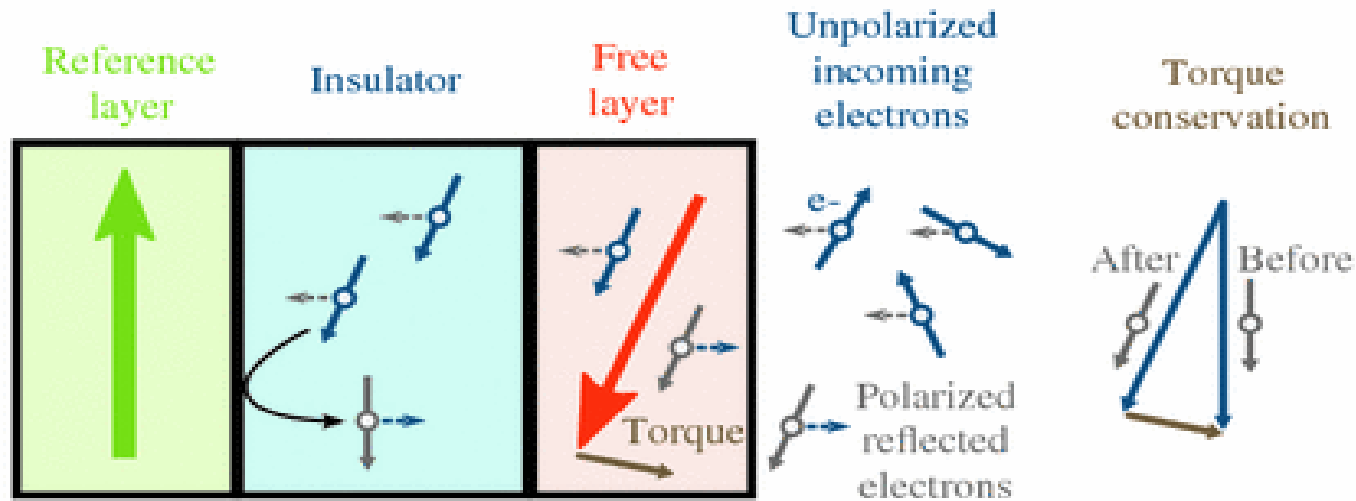
1. Incoming electrons:

- Unpolarized electrons come from the right (entering first the free layer).

2. Partial reflection:

- At the free layer/insulator interface, some electrons are reflected back.
- These reflected electrons become **spin-polarized** according to the free layer's magnetization direction.

Magnetic RAM (MRAM)-Write



(b) Spin-transfer torque on the free ferromagnetic layer, due to the reflected electrons (negative current).

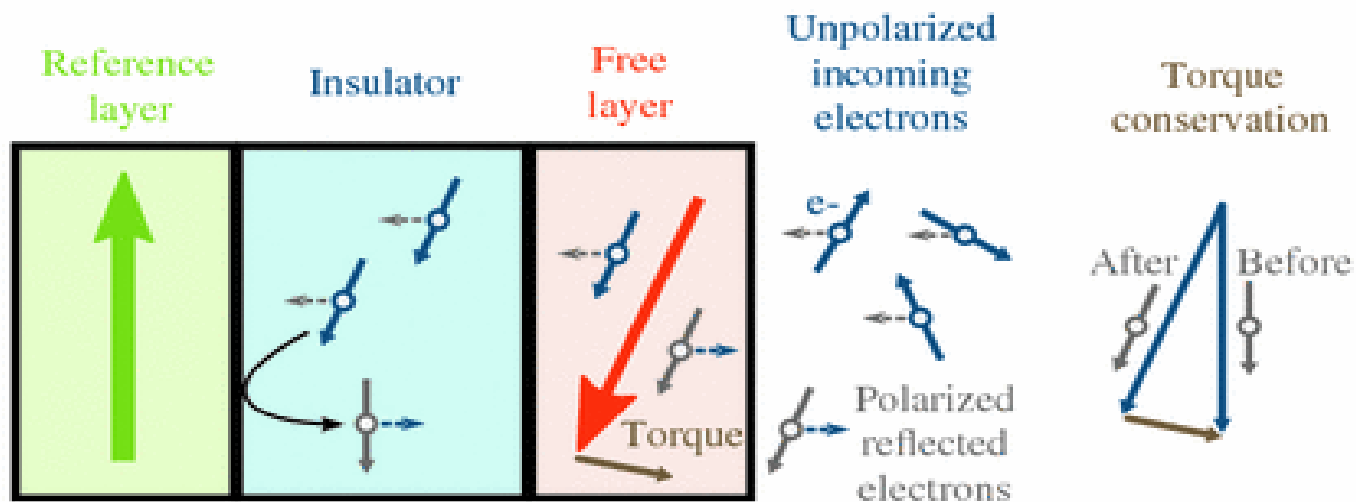
3. Angular momentum exchange:

- During reflection, angular momentum is again transferred—now from the electrons to the free layer, but with **opposite sign**.
- This induces a torque that tends to **rotate the free layer magnetization away** from the reference layer magnetization.

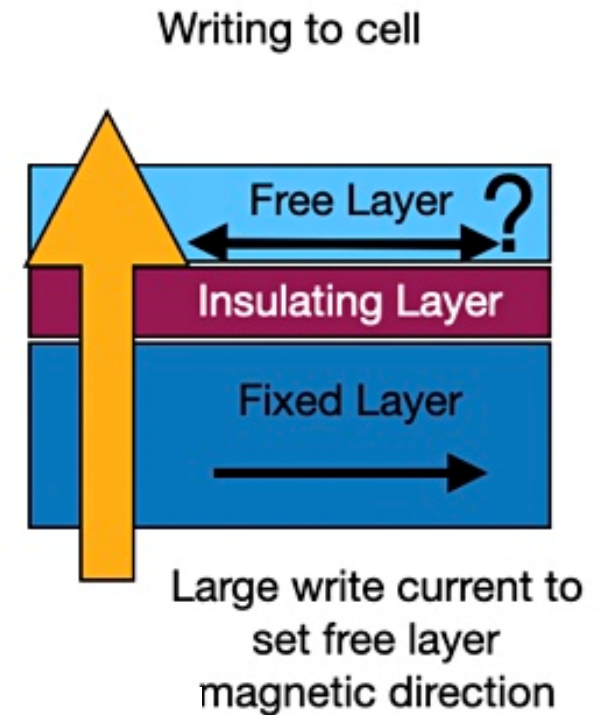
4. Torque conservation:

- As before, the total spin angular momentum is conserved.
- The reflected electrons' spins tilt oppositely, producing the opposite torque direction.

Magnetic RAM (MRAM)-Write

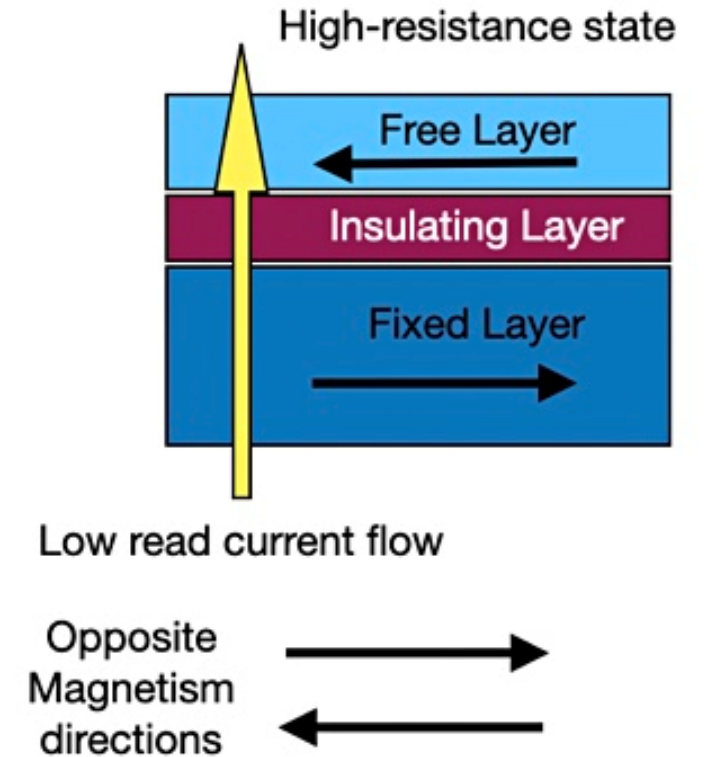
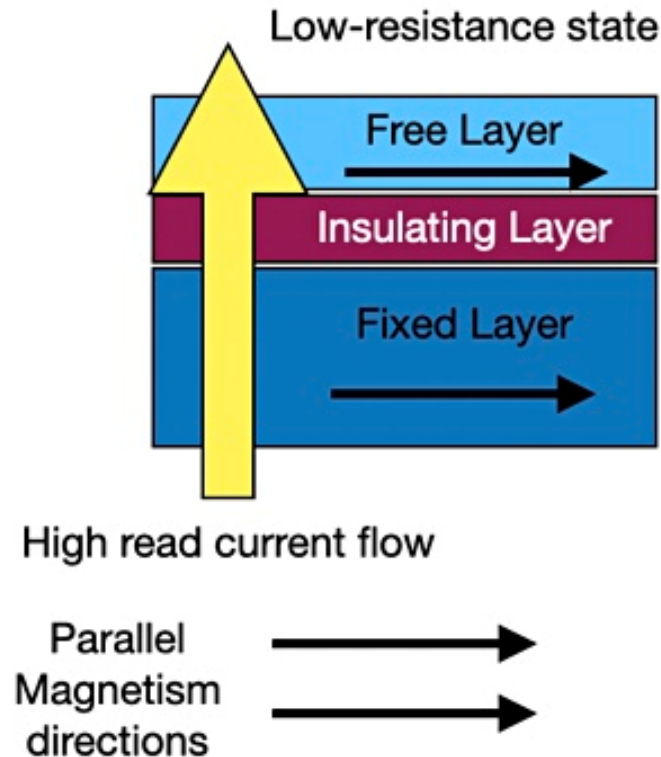
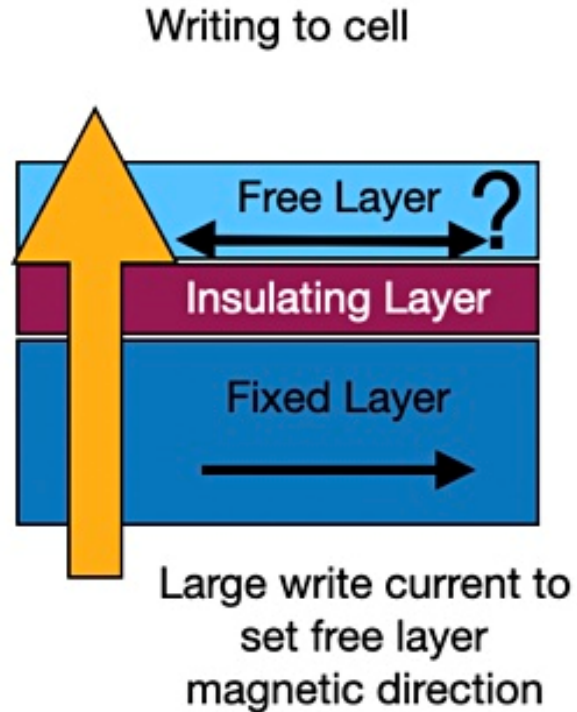


(b) Spin-transfer torque on the free ferromagnetic layer, due to the reflected electrons (negative current).

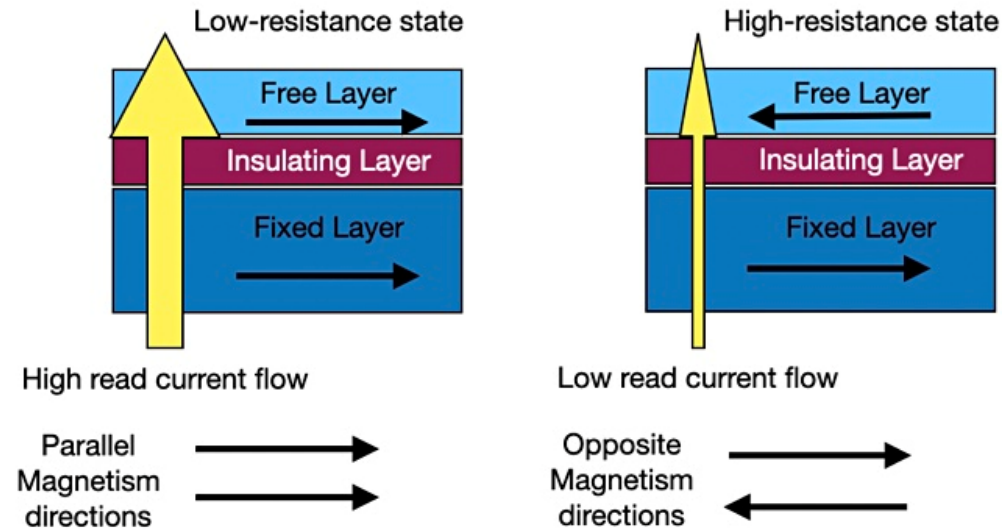


The phase shift of an electron upon reflection is a consequence of the **boundary conditions** that the electron's wave function must satisfy at the interface between two different materials or potentials. This phenomenon is a general property of all wave types, not specific to just electrons. [🔗](#)

Magnetic RAM (MRAM)-Write/Read



Magnetic RAM (MRAM)-Write/Read



Magnetizations are **opposite**.

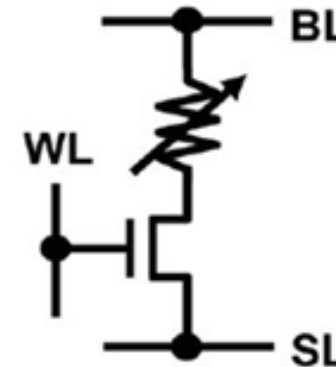
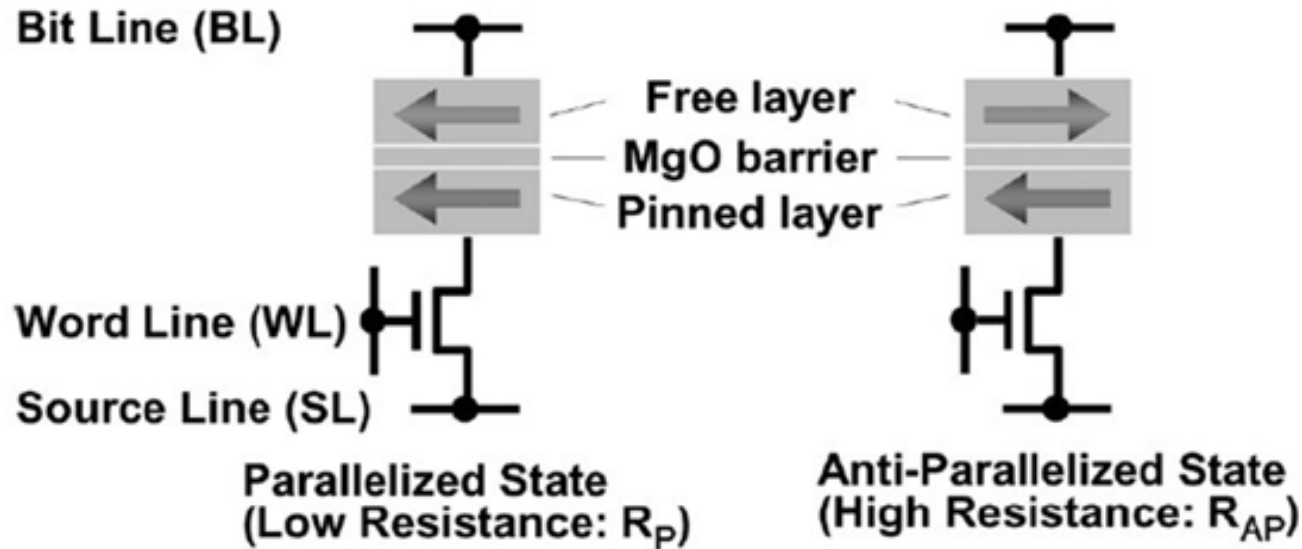
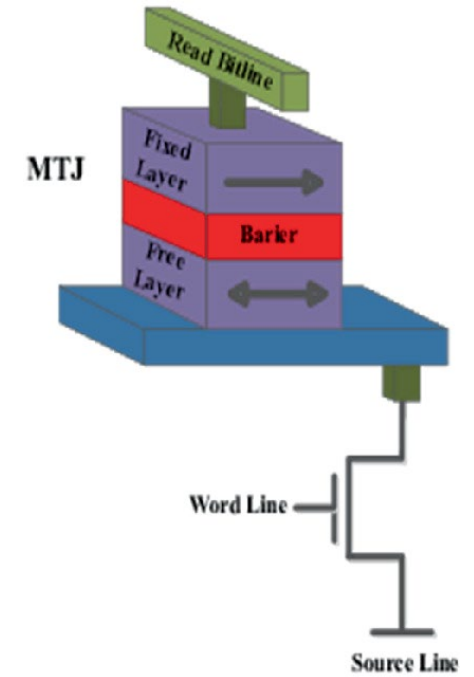
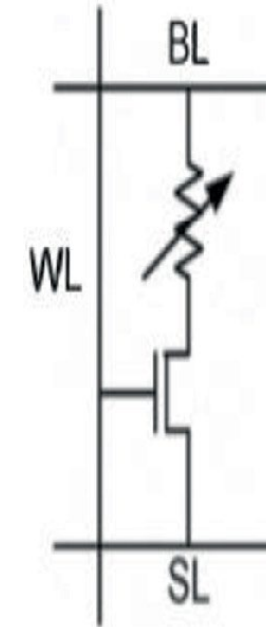
- Majority-spin electrons from the fixed layer ($\uparrow$) now encounter **minority-spin states** in the free layer ($\downarrow$) — which are much fewer.
- Hence, overall tunneling probability drops sharply.
Result $\rightarrow$ **smaller tunneling current, high resistance**.

Magnetizations of both layers are **in the same direction**.

- Majority-spin electrons in the fixed layer (say spin $\uparrow$) find **available majority-spin states** in the free layer.
- Therefore, tunneling is efficient for both spin channels.
Result $\rightarrow$ **larger tunneling probability, higher current, low resistance**.

Magnetic RAM (MRAM)

- ◆ **STT = Spin-Torque-Transfer**
 - Use current to flip free layer between two resistive states

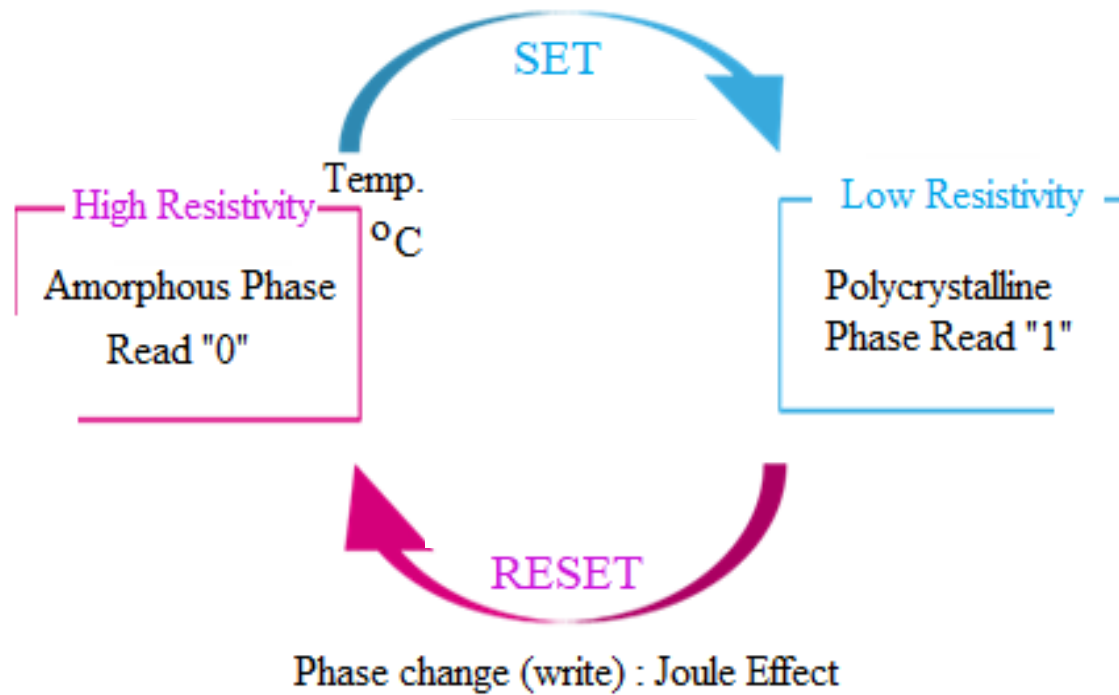


Magnetic RAM (MRAM)

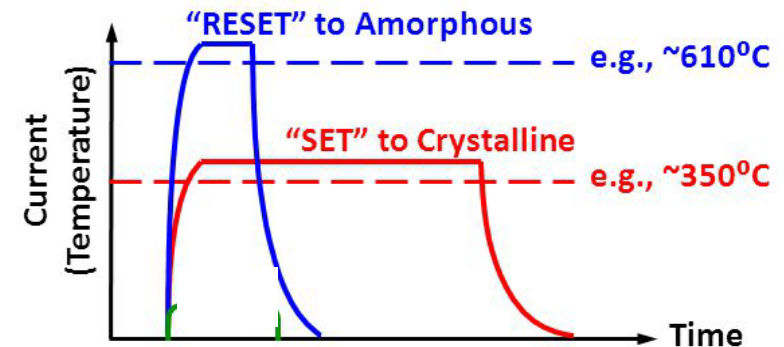
- Everything volatile currently has
 - ▣ High speed (SRAM)
 - ▣ Density (DRAM)
 - AND Everything non-volatile presents
 - ▣ Non-volatility
 - ▣ Low Power (Flash)
 - ▣ Reliability (Hard-drive)
 - On its own
 - ▣ CMOS-compatible
 - ▣ Good scalability potential (over tradition MRAM)
-
- Its **write current, thermal stability, variability, and integration complexity** remain the main bottlenecks.
 - Reducing switching current and enabling CMOS-compatible, scalable MRAM for next-generation embedded memory.

OPTIONAL TOPICS

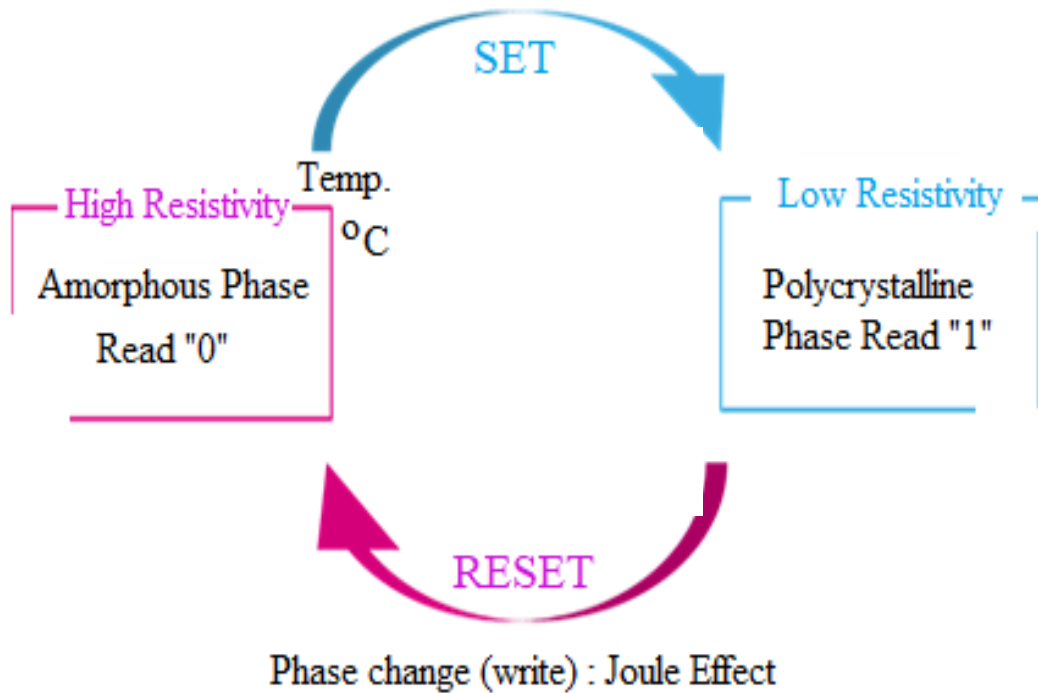
Phase change RAM (PC-RAM)



- Two states of phase change material:
 - **Amorphous:** high resistance, representing "0"
 - **Crystalline:** low resistance, representing "1"
- Operations:

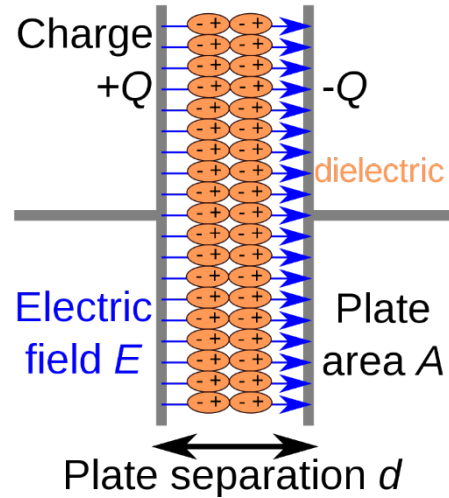


PC-RAM

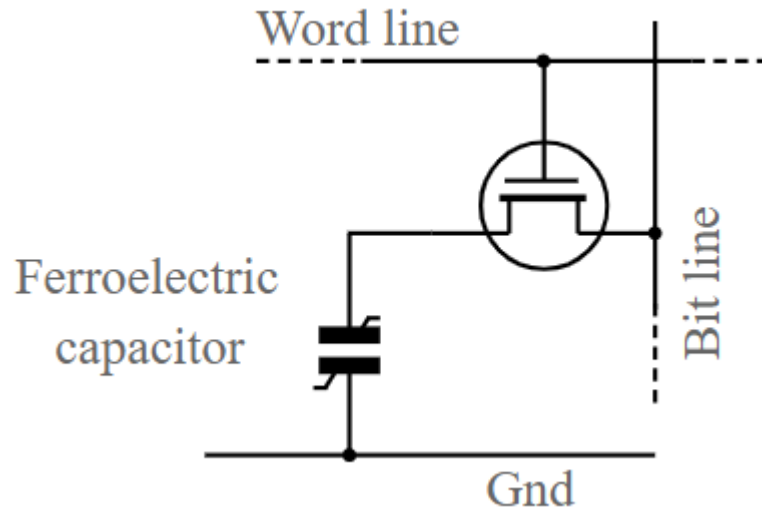


- Is a type of non-volatile random-access memory.
- In PC-RAM, heat produced by the passage of an electric current through a heating element, making it amorphous, or to hold it in its crystallization temperature range.
- PRAM's temperature sensitivity is perhaps its most notable drawback, one that may require changes in the production process of manufacturers incorporating the technology.

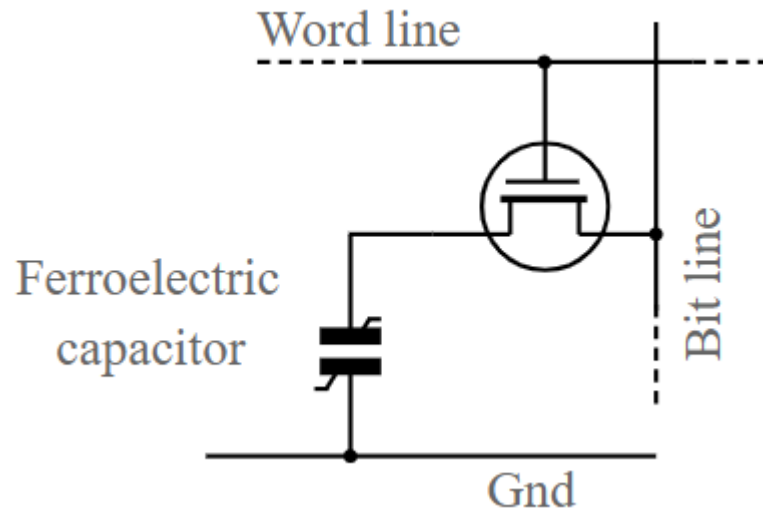
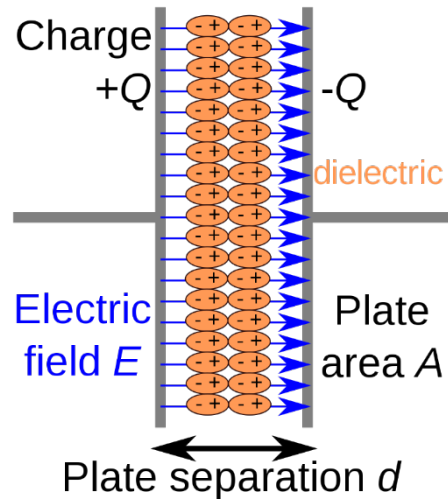
Fe-RAM



- Fe-RAM is similar in construction to the storage cell used in DRAM that both cell types include one capacitor and one access transistor.
- The 1T-1C storage cell design
- In a DRAM cell capacitor, a linear dielectric is used, whereas in a Fe-RAM cell capacitor the dielectric structure includes ferroelectric material, typically lead zirconate titanate (PZT).



Fe-RAM

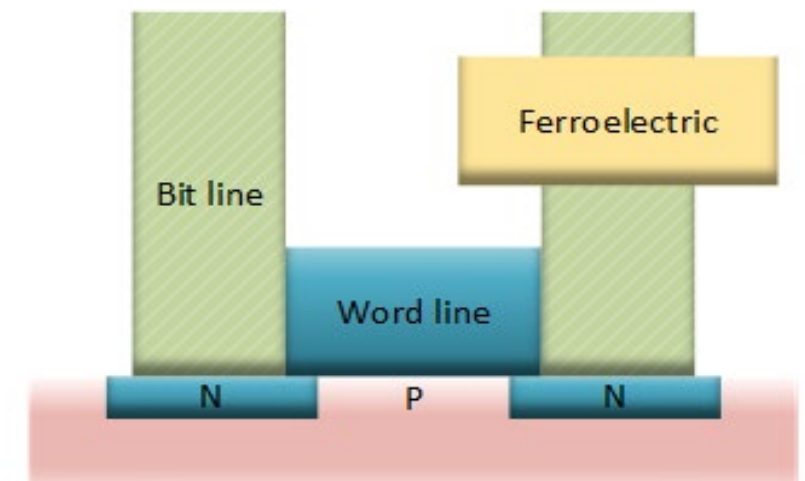
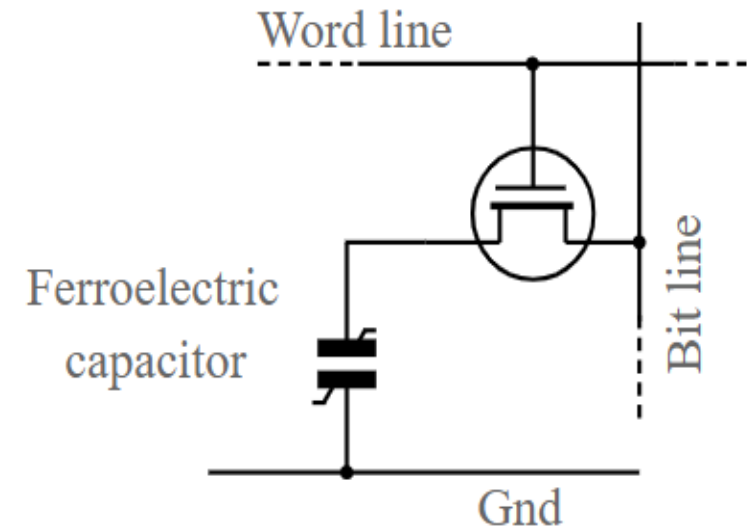


- When an external electric field is applied across a dielectric, the dipoles tend to align themselves with the field direction
- After the charge is removed, the dipoles retain their polarization state. Binary "0"s and "1"s are stored as one of two possible electric polarizations in each data storage cell
- For example, "1" can be encoded using the negative polarization "-Pr", and a "0" can be encoded using the positive polarization "+Pr".

Fe-RAM

Writing:

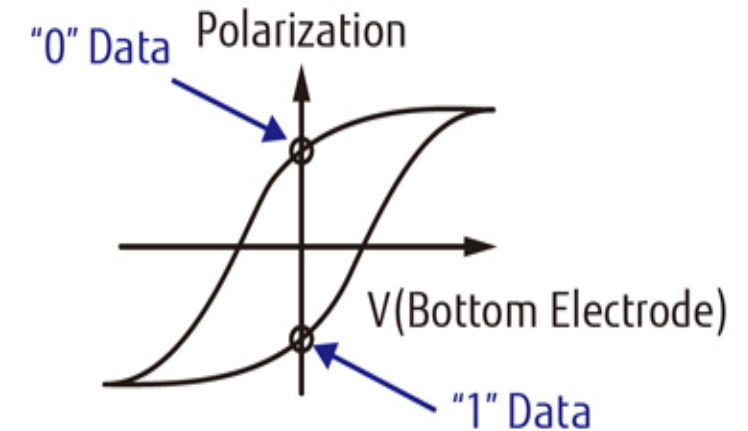
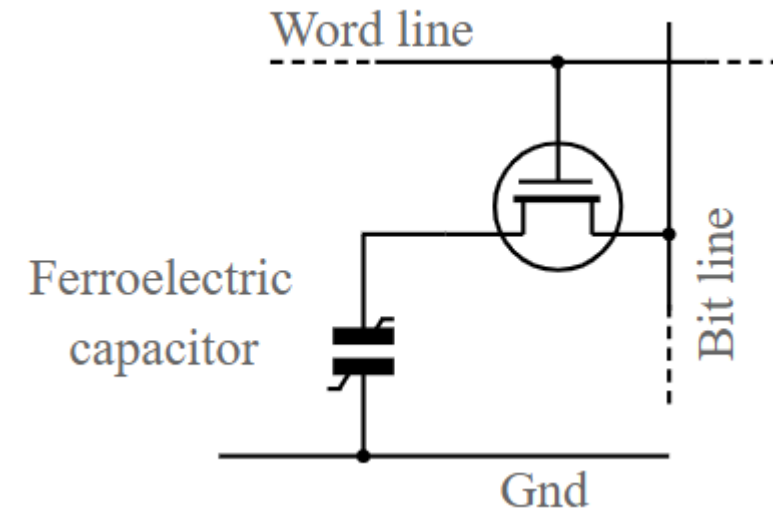
- Writing is accomplished by applying a field across the ferroelectric layer.
- By charging the plates on either side of it, forces the atoms inside into the "up" or "down" orientation (depending on the polarity of the charge), thereby storing a "1" or "0".



Fe-RAM

Reading

- Reading, however, is somewhat different than in DRAM. The transistor forces the cell into a particular state, say "0". If the cell already held a "0", nothing will happen in the output lines.
- If the cell held a "1", the re-orientation of the atoms in the film will cause a brief pulse of current in the output as they push electrons out of the metal on the "down" side.
- The presence of this pulse means the cell held a "1". Since this process overwrites the cell, reading Fe-RAM is a destructive process, and requires the cell to be re-written.



Comparison of Solid-State Memory Technologies (2)

	FeRAM	RRAM	Magnetic field write MRAM	PRAM	STT MRAM
Non-volatile	Y	Y	Y	Y	Y
Memory cell factor (F^2)	16-32	4-6	16-32	5-8	5-7
Read time (ns)	20-50	10-20	3-20	5-20	3-15
Write/erase time (ns)	50	20	10-20	>30	3-15
Number of rewrites	10^{12}	10^5	10^{15} min	10^{12}	10^{15} min
Power consumption at write	Low	Low	Somewhat high	Low	Low
Required input voltage (V)	2-3	1.2	3	1.5-3	1.5